

Reconstruction-independent moment tests for quantum-computed dynamical spectra

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Quantum processors compute dynamical response functions by sampling subspaces that classical resources diagonalize and reconstruct into a spectral function $A(\omega)$ with no independent cross-check. We develop and calibrate a transportable, reconstruction-independent consistency test for such functions. It compares $A(\omega)$ against its own spectral moments $m_k = \int \omega^k A(\omega) d\omega$, which for a zero-temperature Lehmann spectrum are static ground-state expectations ($m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$). Each moment is estimated from the same computational-basis samples that produced the spectrum, not read back off $A(\omega)$, so it can disagree with the reconstruction and refute it. One moment is evaluated on hardware. On retained `ibm.fez` counts of an $L = 8$ addition sector, the coverage moment $m_0 = \langle cc^\dagger \rangle$ has a residual Δ_0 that grows from 0.004 to 0.37 as support coverage falls from 94% to 35%. This exercises the diagonal support-completeness channel only; the discriminating off-diagonal moments m_1, m_2 are evaluated in *simulation* and shown presently infeasible on hardware (a depolarizing bias ~ 37 –115% of the signal). In a blinded, pre-registered test the method caught every determinant-truncation error at a 0.9% false-positive rate. Among spurious-feature artifacts it caught 93% (25/27) of the total-weight-changing ones, already reachable by the coverage channel, and 59% (20/34) of the weight-preserving ones, reachable only off-diagonally; moment-preserving under-converged-Krylov reconstructions pass by construction. As a necessary-condition layer it composes with the sufficient-condition certificates of Wang *et al.* and Mortimer *et al.*, and a closed-form Christoffel bound quantifies the residual. All demonstrations are at classically reproducible sizes; this is the methodological complement of a companion study whose sample-based-diagonalization spectra it reprocesses.

I. INTRODUCTION

Dynamical response functions are the observables that connect many-body theory to experiment: the single-particle spectral function $A(k, \omega)$ is resolved by angle-resolved photoemission [3], the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ by inelastic neutron and resonant x-ray scattering, and the optical conductivity by infrared spectroscopy [4–8]. A quantum processor that computes these observables—rather than the static ground-state energies of the first generation of quantum simulation [9–13]—is producing the functions that are directly measured in correlated-electron experiments, via sample-based diagonalization and quantum Krylov methods on superconducting and trapped-ion hardware [1, 14–18]. Their scientific value is greatest exactly where a trusted classical cross-check is unavailable—the regime that motivates the quantum resource, and whose boundary is actively contested as once-“advantageous” demonstrations are dequantized by improved classical methods [19–24]; the classical-simulability frontier for sample-based diagonalization specifically, and the generative configuration-recovery methods that probe it, are surveyed in [25]. That regime is the eventual target, not the setting of this paper: a screen intended for deployment where no independent standard exists must first have its method fixed and its operating characteristics calibrated where ground truth *is* available, and supply-

ing that method-and-calibration layer—explicitly staged before the field test—is our aim [26–31]. A computed spectrum earns the standing of physical knowledge only once it can be tested against the constraints that every true spectral function of a Hamiltonian must satisfy irrespective of how it was obtained—the moment sum rules that fix its integrated weight and mean frequency [2, 32, 33], the conserved-charge identities its state cannot violate [34], and the positivity of the zero-temperature Lehmann measure [35]: not diagnostics of a particular algorithm, but the physics that survives the method.

A quantum-computed spectrum increasingly sits at the output end of a quantum-centric supercomputing pipeline, in which the device supplies the samples and subspaces that classical resources then diagonalize and reconstruct [1, 15, 36]. A distinct architectural gap motivates the present work. These pipelines increasingly ship a reconstructed spectral *function*— $A(k, \omega)$, $S(q, \omega)$ —with no independent cross-check on the function itself (Fig. 3): a wrong reconstruction can be indistinguishable from a valid one both by visual inspection and by a total-weight check, yet be refuted by a reconstruction-independent moment. The number- and state-certifying guarantees of self-verifying state preparation [37], single-qubit-measurement state certification [38], and reference-free observable estimation [26] do not supply such a check on the reconstructed function. This paper supplies it: a transportable, reconstruction-independent necessary-condition test for the spectral function, attachable by a downstream consumer of quantum-generated spectra—from the sample-

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Two moment branches from the same samples

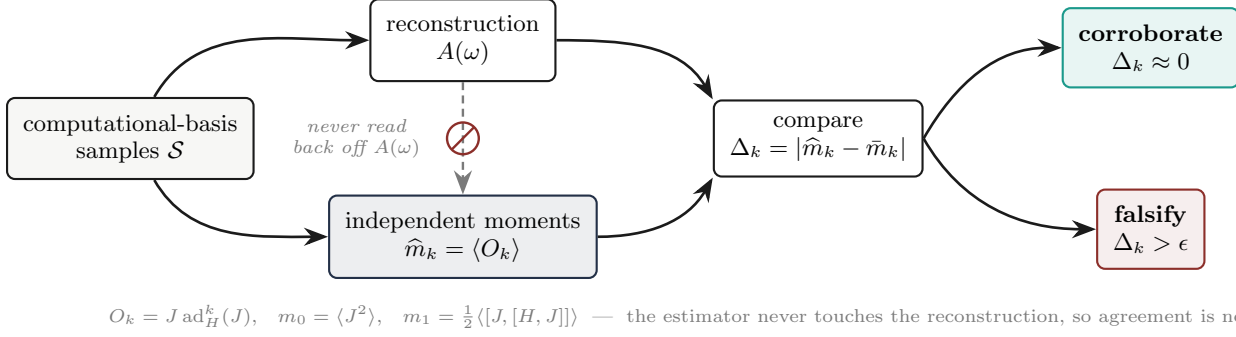


FIG. 1. **Two moment branches from the same samples.** Computational-basis samples feed two branches that must agree: a spectral reconstruction $A(\omega)$ and a reconstruction-independent operator-moment estimator $\hat{m}_k = \langle O_k \rangle$, with $O_k = J \text{ad}_H^k(J)$ the probe times its k -fold commutator with H ($m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$; Appendix A). A discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$ beyond estimator uncertainty refutes the spectrum, where $\bar{m}_k = \int \omega^k A(\omega) d\omega$ is the moment read off the reconstruction. The estimator branch does not read from $A(\omega)$, so it can disagree with the reconstruction, which is what supplies the discriminating power [1, 2].

based-diagonalization and Krylov pipelines treated here to other on-device spectral methods now appearing [39, 40]—and evaluated on the same samples that produced the spectrum [1]. The central methodological claim is that an independent, same-sample moment estimator—pre-registered and blind-tested at a real shot budget—catches the characteristic subspace-truncation and spurious-feature errors of sample-based-diagonalization spectra that a circular check reading moments back off the reconstruction cannot. We position this test within the emerging verification stack for quantum-computed properties as its *necessary*-condition layer, and we anchor the composition on the one object both layers share: the ground-state moment sequence $\{m_k\}$, $m_k = \langle 0|O_k|0 \rangle$. The *sufficient*-condition semidefinite certificates of Wang *et al.* and Mortimer *et al.* solve a feasibility relaxation that returns *two-sided* intervals $m_k \in [\ell_k, u_k]$ on exactly these moments [41, 42]—the same moments our screen estimates independently off the samples—so their feasibility *certifies* while our infeasibility *refutes*: a sampled or read-off $\{m_k\}$ that no positive measure carries, or that misses the certified interval, rejects the reconstruction. Because our screen reads those moments from a frequency-resolved Lehmann *measure* rather than a static expectation value, the composition necessary $\circ$ sufficient—three conditions on one $\{m_k\}$ —is well defined on a single moment sequence; we make it concrete at the exact-moment level in Sec. IX A.

Two influential responses to this predicament already exist, and it is essential to delineate them by mechanism rather than by aspiration. Self-verifying variational simulation prepares a target state variationally on the device [10, 43, 44] and certifies it by measuring an independent Hamiltonian-derived quantity—an energy variance—on that same state, bounding its distance to a true eigenstate [37]. A distinct line of work

establishes trust in error-mitigated observable estimates by validating the hardware noise model and the mitigation procedure [45–49], or by predicting many observables from randomized measurements without full tomography [50, 51], without recourse to a classical reference [26]. Both approaches certify a *number*—an energy or an expectation value—and both are powerful within that remit [26, 37]. Neither is designed to interrogate the internal self-consistency of a reconstructed *function* such as a full $A(\omega)$, and neither screens the characteristic failure mode of subspace-based spectral methods, in which spectral weight is placed at the wrong frequencies while integrated quantities can remain within tolerance—a mode now documented in critical studies of quantum-selected configuration interaction and quantum subspace diagonalization [1, 52–56], and independently motivating response-targeted determinant selection on the classical side [57].

We treat the checks as necessary conditions and attach to the spectrum a battery of them—tests the true spectrum *must* pass—so that a reconstruction that fails any is refuted, while passing them is corroboration, not proof. The value is precise and modest: a spectrum equipped with the test is one that could have failed a real test and did not.

The moments tested are the spectral moments $m_k = \int \omega^k A(\omega) d\omega$. For a zero-temperature Lehmann spectral function these moments coincide with static ground-state expectation values— $m_0 = \langle J^2 \rangle$, $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$, and so on—obtainable directly from operator algebra rather than from the dynamics [35, 58]. The double-commutator form of these moment/sum-rule identities is standard [2].

Using such moment sum rules to *check* a computed spectrum is itself long-standing: in the dynamical-mean-field and quantum Monte Carlo literature the low-order

spectral moments of the retarded Green’s function and self-energy have served as internal consistency tests on numerically continued spectra, developed most systematically in the sum-rule program of Turkowski and Freericks for the Falicov–Kimball and Hubbard models [59, 60]. It is equally standard practice in the analytic-continuation, maximum-entropy, and stochastic-continuation pipelines to report low-order sum-rule agreement as an internal consistency check on a continued spectrum [61, 62]; we claim no novelty for the use of moment sum rules to test a spectrum *per se*. What is new is not the identities, nor the bare act of checking them, but four specific and separable choices in how they are deployed. First, each moment is estimated *independently* of the reconstruction—as an operator expectation, never read back off the continued $A(\omega)$. Against a trusted classical reference this independence is already standard practice in the sum-rule and analytic-continuation traditions [59–62]; what is new here is that the estimate is taken from the *same device samples* that produced the spectrum—inheriting their shot noise and mitigation bias—so it can disagree with the reconstruction and falsify it precisely in the regime where no classical reference exists, rather than merely annotate a spectrum the same data produced. Second, the battery and its falsification thresholds are fixed by a blinded pre-registration before any instance exists, so no threshold is tuned to the outcome it judges; this control transfers to any test whose thresholds can be fixed in advance. Third, the screen is positioned as the *necessary-condition* layer of a verification stack, composing with the sufficient-condition property certificates of Wang *et al.* and Mortimer *et al.* (necessary $\circ$ sufficient) but acting on a frequency-resolved Lehmann measure rather than a static observable [41, 42]. Fourth, the classical Christoffel–Darboux and super-resolution machinery is deployed to *quantify* the screen—the cumulative-weight miss-distance (Sec. VII) and the detectability threshold (Sec. VIII)—rather than to prove new mathematics. The moment identities, the Christoffel bound, and the Vandermonde-conditioning results are all classical and credited as such; what is new is only the reconstruction-independent same-device *sampling* of the moments, the pre-registered protocol, their composition into a necessary-condition verification layer, and the Christoffel/detectability deployment that calibrates it. To our knowledge no prior method combines these—same-sample estimation off the device, blinded pre-registration, and necessary-condition composition with sufficient-condition certificates—and it is this combination, not any single ingredient, that is the contribution (Table I).

In the optical-conductivity setting $m_0 = \langle J^2 \rangle$ is the total integrated current–current weight, whereas the optical f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$ is the distinct m_{-1} moment—a ground-state kinetic energy whose character was established by Kohn and made explicit for the Hubbard chain by Maldague [4, 32, 33]. The question of which moment sequences $\{m_k\}$ are consistent

with a positive spectral measure, and of how tightly finitely many moments constrain that measure, is the classical moment problem, with a mature theory of existence, determinacy, and extremal bounds [66–69]. This same moment-problem machinery has very recently been turned into rigorous, reconstruction-independent bounds on smeared matrix-valued spectral functions in Euclidean lattice field theory by Abbott, Jay, and Oare [63] and, in a closely related two-sided semidefinite form, by the causal-bootstrap program of Abbott *et al.* [64] and by Mutzel and Tilloy [65]; our screen differs in object and in use—it estimates the moments from real-time, post-selected device counts of a Krylov–Trotter flow and applies an operational falsification threshold, rather than bounding a reconstruction of Euclidean-time correlators—and we claim no primacy for the moment problem as a source of reconstruction-independent constraints. The bridge from moments to orthogonal polynomials and Gauss quadrature—the machinery underlying our closed-form bound—is likewise classical [70–74], and is the same machinery that powers the kernel-polynomial and recursion methods routinely used to compute spectra on classical hardware [75–78].

The central design choice is *independence* (Fig. 1): each moment is evaluated as an operator expectation value on the same computational-basis samples, not by integrating the reconstructed $A(\omega)$ [1]. A moment read back off the reconstruction is circular—guaranteed to agree, and therefore unable to refute—whereas an independent estimator built from operator algebra can disagree with the reconstruction and thus flag an error, confronting the spectrum with quantities it did not itself generate.

We exercise the screen across error classes and physical domains. The quantitative falsification here is carried by the current-response probe and the determinant-truncation study; the breadth across observables— $A(k, \omega)$, $S(q, \omega)$, $S^{zz}(q, \omega)$ —rests on the shared moment structure that every $T = 0$ Lehmann channel inherits, so the screen is transportable by construction rather than tuned to one probe. In the doped Hubbard current response, a reconstruction that introduces a spurious low-frequency (Drude-like) peak, misplacing mid-infrared weight to low energy, preserves the total integrated weight $m_0 = \langle J^2 \rangle$ yet fails the first current moment m_1 (the mean frequency once normalized by m_0), exposing a wrong distribution of a correct total weight [32, 33]. In sample-based diagonalization, determinant-subspace truncation is detected by the independent moment estimator while a back-computed control remains blind—a clear demonstration of why circularity must be avoided [14, 15]. We then subject the screen to a blinded, pre-registered test whose battery and thresholds were sealed by hash before any instance existed: it catches every support-truncation error and most spurious-feature artifacts at a low false-positive rate, while missing—by construction—the moment-preserving reconstructions it is designed-

TABLE I. **Contribution relative to the closest prior art.** Each neighboring method supplies one ingredient; the present test is distinguished by combining same-device-sample moment estimation (correlated with the reconstruction’s own shot noise and mitigation bias), blinded hash-sealed pre-registration of thresholds, and a *necessary*-condition role that composes with the *sufficient*-condition moment certificates. “—” marks an axis not applicable to that method’s object. For this work the on-device entry is the diagonal coverage moment m_0 (`ibm_fez`, $L = 8$); the discriminating off-diagonal moments m_1, m_2 are evaluated in simulation and deferred to lower-error hardware (Table II).

method	recon.-indep.	on-device	blinded	role
Turkowski–Freericks sum-rule checks [59, 60]	yes (vs classical)	no	no	consistency check
maxent / stochastic continuation [61, 62]	yes (vs classical)	no	no	consistency check
Wang–Acín / Mortimer SDP [41, 42]	— (bounds moments)	no	no	sufficient
Abbott <i>et al.</i> / causal bootstrap [63–65]	yes (Euclidean)	no	no	two-sided bound
Kokail <i>et al.</i> self-verification [37]	— (certifies state)	yes	no	sufficient (state)
reference-free observable estimation [26]	— (certifies number)	yes	no	validity
this work	yes	yes (m_0)	yes	necessary

blind to. As an orthogonal companion—a state-level identity, not a moment screen—the exact Gauss-law constraint $[H, G_x] = 0$ catches gauge-symmetry breaking in a $U(1)$ lattice-Schwinger model, which we develop in Appendix D [34, 37]. We run the reconstruction-independent coverage moment on real IBM Heron data: from retained `ibm_fez` counts of a purpose-built $L = 8$ addition-sector job its residual Δ_0 grows as support coverage falls (Sec. VI), a device-side demonstration on the diagonal support-completeness channel, an earlier full-coverage $L = 6$ run [1] having established executability. We close with a closed-form bound: under exact moments the extremal spread in cumulative spectral weight equals the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ at interior points, a computable bound that contracts monotonically as more moments are enforced [70, 79]. We give the falsifying side an equally closed form: a completeness theorem (Theorem 2) characterizing exactly which reconstruction errors the moment battery detects—on N distinct poles the first N moments are complete for fixed-pole weight redistributions—and tying the power to falsify the remainder to sample complexity through the Vandermonde super-resolution limit, so the residual blind spot is not a gap in the method but a quantified shot-cost that diverges only as reconstructed poles merge.

We are deliberate about scope from the outset. The method falsifies but does not certify; its conditions are necessary but not sufficient. The moment screen targets the determinant-subspace-truncation and spurious-feature error classes of sample-based-diagonalization (SQD/QSCI) spectra—weight and subspace misassignment in a reconstruction; a complementary exact operator identity (the Gauss law) targets gauge-symmetry violation in the prepared state. It is, by the same token, *out of scope* for Krylov-convergence errors: an under-converged quantum-Krylov or time-evolution reconstruction reproduces the low-order moments by construction and therefore passes the battery (Sec. V), and detecting that error class calls for a separate instrument—most naturally a Ritz/residual-norm convergence check on the

Krylov basis—which we leave to future work. Neither is a general device-noise diagnostic, which remains the province of noise-model and observable-estimation validation [26]. It certifies integrated weight, not pointwise line shapes; its positivity structure is that of a diagonal $T = 0$ Lehmann spectrum [35]; and its exact-moment idealization must be softened to robust interval moments before it is trustworthy on noisy hardware.

II. METHOD: RECONSTRUCTION-INDEPENDENT MOMENT TESTS

We formalize the screen in two steps: first the object it tests—a quantum-computed spectrum and the samples that produced it—and then the falsifiers we attach to it.

A. The quantum processor and the origin of the samples

The object the screen tests is produced by a definite quantum-computational pipeline, and the screen acquires meaning only once that pipeline is fixed. The Krylov–Trotter pipeline and the full-coverage $L = 6$ spectral reconstruction reprocessed here are those of the companion study [1]; the present paper is its methodological complement—it contributes the reconstruction-independent test, its blinded calibration, its closed-form bounds, and a new purpose-built $L = 8$ `ibm_fez` coverage job (Sec. VI), not the companion’s spectra. The processor’s task, inherited from the companion study [1] and its sample-based-diagonalization lineage [14–16], is not to evaluate $A(\omega)$ directly but to *define, by physical sampling, the correlated subspace* in which the spectrum is then built by exact classical diagonalization. A correlated input state is prepared on an IBM Heron processor—a hardware-native local unitary cluster-Jastrow ansatz in the chemistry setting [15], or a Trotterized real-time

evolution of an $(N+1)$ -particle seed $c_p^\dagger|\psi_0\rangle$ in the dynamical setting of the companion study [1, 80, 81]—and measured shot by shot in the occupation-number basis. Each shot returns a bitstring—one candidate Slater determinant—and its empirical frequency across shots estimates $|\langle\text{config}|\psi\rangle|^2$ [14]. The distinct measured bitstrings, after symmetry-restoring configuration recovery to the correct particle-number and spin sector [25], form the subspace $\mathcal{S}$; the Hamiltonian is projected onto $\mathcal{S}$ and diagonalized classically, and the dynamical spectral function is assembled as the $T = 0$ Lehmann sum $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta_\eta(\omega - (E_n - E_0))$ over the resulting eigenpairs, with δ_η a narrow broadening [1, 35].

At scale the processor's role is essential—though not at the classically reproducible sizes demonstrated here, where the full sector is recovered by coverage. The exact many-body sector grows exponentially with system size, so the configurations carrying spectral weight cannot in general be enumerated classically. The correlated prepared state instead acts as a physically informed importance sampler, concentrating shots on exactly those configurations, so that $\mathcal{S}$ is fixed by measurement rather than by a classical selection heuristic [1, 15]. This is precisely what makes the screen non-trivial rather than circular: the reconstruction-independent moment estimators $\hat{m}_k = \langle O_k \rangle$ defined next are evaluated on the *same* measured bitstrings $\mathcal{S}$, and inherit the same shot noise and error-mitigation choices as the spectrum they test.

B. The independent moment estimator and the falsifier

A quantum-computed dynamical spectrum is thus delivered as a reconstructed non-negative function $A(\omega)$ together with the sample set $\mathcal{S}$ just described [1]. For a zero-temperature diagonal Lehmann spectral function of an operator J , $A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0))$ is a positive measure on the real line, and its power moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectation values of operators built from nested commutators of J with H : $m_0 = \langle 0|J^2|0\rangle$, $m_1 = \frac{1}{2}\langle 0|[J, [H, J]]|0\rangle$, and in general $m_k = \langle O_k \rangle$ with $O_k = J \text{ad}_H^k(J)$ the probe times its k -fold commutator with H , derived in Appendix A [2, 35]. This is the standard moment/sum-rule construction [2]; in the optical case $m_0 = \langle J^2 \rangle$ is the total integrated weight and m_1 the first moment (the mean frequency once normalized by m_0), while the optical f -sum is the separate m_{-1} moment [32, 33].

The screen attaches to the spectrum an *independent* estimator $\hat{m}_k$ of each moment. Independence has an operational definition: $\hat{m}_k$ is computed as the sample estimate of $\langle O_k \rangle$ over $\mathcal{S}$ —via the same diagonal-in-the-computational-basis or importance-weighted estimators used to obtain ground-state observables in sample-based diagonalization—and is at no point obtained by numerically integrating $\omega^k A(\omega)$ [1, 14, 15]. We denote by $\bar{m}_k = \int \omega^k A(\omega) d\omega$ the reconstruction moment, the

quantity a circular check would use. The falsifier is the discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$: the true spectrum must satisfy $\Delta_k = 0$ up to statistical error, so a discrepancy exceeding the estimator's uncertainty is a decisive refutation. Crucially, a circular check compares $\bar{m}_k$ to itself and yields $\Delta_k \equiv 0$ by construction, and therefore can never fire; Results II demonstrates this contrast on controlled truncation.

The screen is a battery, not a single test: it enforces Δ_k for $k = 0, 1, \dots, K$ and, separately, the moment-problem consistency of the sequence $\{\hat{m}_k\}$ itself—the Hankel positive-semidefiniteness that any sequence of moments of a positive measure must satisfy [66, 67]. A violated Hankel condition falsifies the spectrum even without reference to $A(\omega)$ —a moment sequence is thus refutable by geometry alone (Fig. 2)—because no positive measure, hence no physical $T = 0$ Lehmann spectrum, has those moments [67, 68]. Because a diagonal $T = 0$ Lehmann measure is supported on $\omega = E_n - E_0 \geq 0$, the sharp condition is in fact the Stieltjes pair $H_n = [m_{i+j}] \geq 0$ and the shifted $H_n^{(1)} = [m_{i+j+1}] \geq 0$; the shifted matrix is an additional native falsifier that a bare Hankel test omits. The battery also accepts any exactly known operator identity as a native falsifier: a conserved charge $[H, G] = 0$ with a known physical eigenvalue supplies a moment-like constraint whose violation is likewise decisive, as we exploit in the gauge-theory domain [34]. The construction is transportable because it depends only on (a) a positive spectral measure and (b) operators whose ground-state expectations are computable from the same samples—both of which hold across the domains we treat [70, 75].

III. RESULTS I: WEIGHT-MISASSIGNMENT ERRORS ACROSS OBSERVABLES

We begin by displaying the physical spectra on which the screen operates. Figure 3 shows the single-particle spectral function $A(k, \omega)$ of the half-filled Hubbard ring, with its lower and upper Hubbard bands separated by the Mott gap and the characteristic dispersion of a one-dimensional correlated insulator [1, 5, 82–85]. A truncated or otherwise corrupted reconstruction misplaces spectral weight. Here the reconstructed Hubbard-band centroid dispersion is flattened by a momentum-resolved inter-band weight misassignment that leaves the Mott gap and the per- k total weight intact—the single-particle, momentum-resolved counterpart of the optical weight-misplacement of Fig. 5 and of the weight-misassignment family quantified in Fig. 7. The reconstruction-independent first moment, evaluated at each momentum k via the anticommutator sum rule appropriate to the non-Hermitian single-particle probe $c_{k\sigma}$ (Appendix A), detects the misplacement (right panel). The same construction applies to the collective response: Figure 4 shows the charge dynamical structure factor $S(q, \omega)$, gapped by a particle-hole (density) excitation

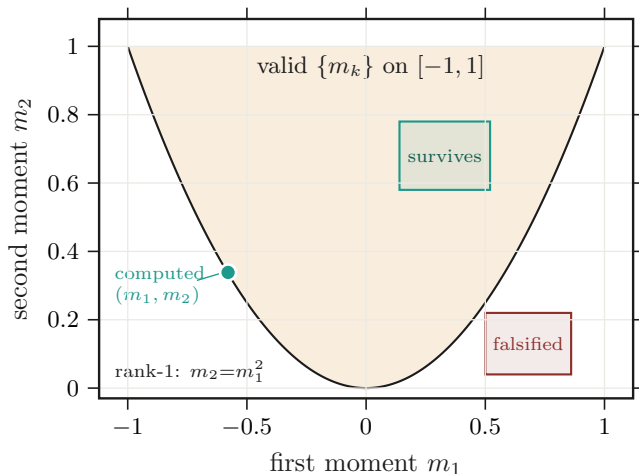


FIG. 2. **Hankel-positivity region in the (m_1, m_2) plane.** In the (m_1, m_2) plane (normalized measure rescaled to $[-1, 1]$), a valid spectrum's moments must lie in the moment body $m_1^2 \leq m_2 \leq 1$ of a measure supported on $[-1, 1]$; the lower boundary $m_2 = m_1^2$ is the Hankel (rank-one) edge, a single δ , while the upper boundary $m_2 = 1$ is the support-localizing (Stieltjes) constraint of the $[-1, 1]$ rescaling. A candidate whose shot-budget interval box lies entirely below the rank-one boundary—wholly in the Hankel-negative region $m_2 < m_1^2$ (oxblood)—is refuted with no reference to $A(\omega)$, whereas one that lies inside the body (teal) survives. The real computed moments of the doped-Hubbard current spectrum (teal dot) sit essentially on the boundary—the measure is dominated by one spectral atom, hence nearly rank-one [66, 67].

onset Δ_c , alongside the spin structure factor $S^{zz}(q, \omega)$, whose gapless low-energy weight is the spinon continuum of the gapless spin sector [86]—the hallmark of spin-charge separation [1, 2, 5–7]. Each channel is a positive $T = 0$ Lehmann measure carrying its own moment sum rules, so the screen transports across observables as well as across models. Such spectra are the benchmark output of mature classical many-body methods—density-matrix renormalization group and tensor-network dynamics, dynamical mean-field theory, and determinant quantum Monte Carlo with analytic continuation—against whose established features the moment screen's necessary conditions can themselves be checked [61, 87–90].

We now exercise the screen on a wrong distribution of correct total weight. In the doped one-dimensional Hubbard model the regular ($\omega > 0$) current spectral function $A_J(\omega)$ that the screen tests is dominated by a mid-infrared/Mott band, and its total integrated current-current weight is fixed by the zeroth moment $m_0 = \langle J^2 \rangle$ —a ground-state quantity [4, 32, 33, 93]. On the finite ring this screened object is *entirely regular*: the charge stiffness is a separate zero-frequency Drude delta that does not appear inside $A_J(\omega)$. We construct a wrong reconstruction that injects a spurious low-frequency, *Drude-like* peak at small nonzero ω , misplacing a fraction of the mid-infrared weight to low

energy while preserving the total. Figure 5 shows the outcome: the zeroth moment m_0 (the total integrated weight) is satisfied to within estimator error, so a total-weight check alone passes the wrong spectrum; the first current moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$, computed independently from the samples, disagrees with the reconstruction's first moment by many times its uncertainty, and the spectrum is refuted [2, 32]. The lesson is that integrated total weight is a weak falsifier and that the first moment—the mean frequency of the weight—is precisely the quantity a misplaced-weight reconstruction corrupts.

The same misplaced weight that the first moment m_1 exposes is exposed even more sharply by the *negative-order* moment, and for a complementary reason: the positive power moments $m_k = \int \omega^k A_J d\omega$ carry the kernel ω^k and so weight *high* frequency, where they are weakest precisely where this error lives, whereas the inverse moment $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega = \sum_n |\langle n|J|0 \rangle|^2/\omega_n$ carries the opposite $1/\omega$ kernel and is maximally sensitive to low-frequency weight (Fig. 5). This inverse moment is the classic optical f -sum rule, and it too is a reconstruction-independent ground-state expectation: since the current is the polarization velocity $J = i[H, P]$, one has $m_{-1} = \frac{1}{2} \langle [P, [H, P]] \rangle = \frac{1}{2} \langle -\hat{T} \rangle$, a ground-state kinetic energy (equivalently the conductivity f -sum $\int_0^\infty \text{Re } \sigma_{\text{reg}} d\omega = \pi m_{-1} = \frac{\pi}{2} \langle -\hat{T} \rangle$), which we verify against the exact Lehmann sum to $\sim 10^{-15}$ on the open chain [2, 32, 33]. On the periodic ring of Fig. 5 the polarization is multivalued and the identity carries the exact charge-stiffness correction Kohn established, $m_{-1} = \frac{1}{2} \langle -\hat{T} \rangle - D$ with $D = \frac{1}{2} \partial_\theta^2 E_0$ the finite-size Drude weight (verified to $\sim 10^{-6}$ against the flux curvature), so on this doped ring the regular spectrum is depleted at low frequency—the coherent weight is the *separate* zero-frequency stiffness D , leaving only 1.4% of m_0 below the mid-infrared band—and $m_{-1} = 0.324$ is correspondingly small. A spurious low-frequency peak is therefore amplified twice over, through the $1/\omega$ kernel and against this small baseline: at the same injected corruptions that leave m_0 blind and move m_1 by up to 38%, the inverse moment moves by up to 154%—a roughly $4\times$ larger relative discrepancy than m_1 (38%), and larger still than m_2 (Fig. 5). The inverse moment is also the *cheapest* of the battery to estimate: $\langle -\hat{T} \rangle$ is a one-body operator, measured from two qubit-wise-commuting settings that are a subset of those the off-diagonal m_0, m_1 estimator already requires (Appendix B), so m_{-1} adds essentially no shot cost while catching the low-frequency error the high-frequency-weighted positive moments respond to only mildly. We deploy it not as a new sum rule—it is the textbook f -sum [32, 33]—but as an independent negative-order test attacking the primary error class from the opposite end of the frequency axis [1].

We now convert the transportability across observables from an assertion into a demonstration, running the same screen on the collective channels of Fig. 4 (Fig. 6). For each collective probe $A \in \{\rho_q, S_q^z\}$, with

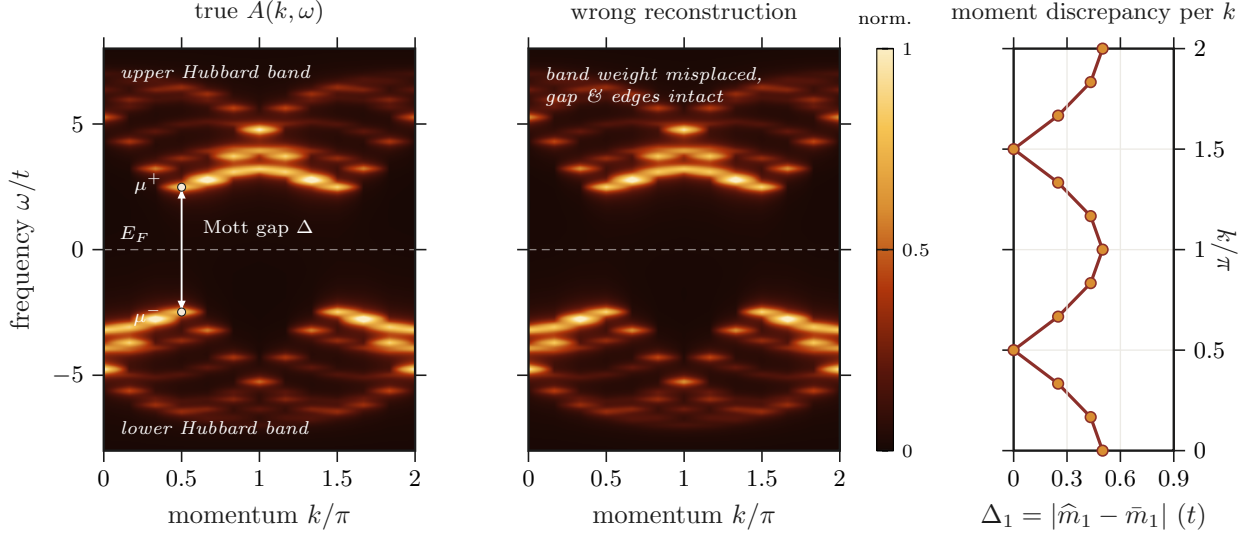


FIG. 3. **The reconstruction-independent test on the single-particle spectral function $A(k, \omega)$ of the Hubbard model.** Left: the true $A(k, \omega)$ of the half-filled Hubbard ring, with the Fermi level E_F at mid-gap and the dispersing upper/lower Hubbard bands (UHB/LHB) separated by the Mott gap; the correlation-driven edges $\mu^\pm \approx \pm 2.484t$, marked by the dots on the band edges at $k_F = \pi/2$ and spanned by the arrow, give the single-particle Mott gap $\Delta = \mu^+ - \mu^- \approx 4.97t$ (approaching U only in the atomic limit $t \rightarrow 0$). Center: an illustrative injected corruption, modeled on the determinant-truncation weight-misassignment mode quantified in Fig. 7: at each momentum a fraction of upper-Hubbard-band (electron-addition) intensity is rebalanced against the lower band, flattening the reconstructed band-centroid dispersion while leaving the Mott gap empty of poles (the reconstructed $A(k, E_F)$ is only the finite- η Lorentzian tail, $\lesssim 1\%$ of the band-edge weight), the edges $\mu^\pm$ fixed, and the per- k total weight fixed by the equal-time anticommutator sum rule $m_0 = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$ preserved by the corruption to machine precision—a Mott spectrum that both a casual inspection and a total-weight check accept. Right: the reconstruction-independent first moment $\hat{m}_1(k) = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U\langle n_{-\sigma} \rangle$, the Hartree-shifted band centroid computed per momentum from the ground state (Appendix A), departs from the reconstruction's $\bar{m}_1$ by $\Delta_1 = |\hat{m}_1 - \bar{m}_1| \propto |\epsilon_k|$: the discrepancy Δ_1 is largest where the bare dispersion $|\epsilon_k|$ is largest, at the band extrema $k = 0, \pi, 2\pi$ ($\Delta_1 = 0.50t$), and vanishes exactly at the native momenta $k = \pi/2, 3\pi/2$ where $\epsilon_k = 0$ —a self-consistency point that no flat-dispersion error can evade. For this single-particle probe m_0 is pinned to 1 by the equal-time anticommutator and carries no discriminating power (blind), so the first moment is the first non-trivial falsifier [1, 2, 35, 60]. Sector-basis Lanczos ground state with Haydock continued-fraction Green's functions [74, 76], $L = 12$ sites at half filling (12 exact momenta), $U/t = 8$, broadening $\eta = 0.16t$. The single-particle spectrum is the canonical locus of spin-charge separation—its spinon and holon branches are here only partially resolved at this size and broadening—while the collective response of Fig. 4 separates the two energy scales more cleanly.

$\rho_q = \sum_j e^{-iqj} n_j$ and $S_q^z = \sum_j e^{-iqj} S_j^z$, the two independent moments are the same ground-state operator expectations as for the current, $m_0(q) = \langle A^\dagger A \rangle_c$ and $m_1(q) = \langle A_0 | (H - E_0) | A_0 \rangle = \frac{1}{2} \langle [A^\dagger, [H, A]] \rangle$; for the non-Hermitian collective probe the last equality is the symmetrized form $\frac{1}{2} \langle [A^\dagger, [H, A]] \rangle = \frac{1}{2} [m_1(q) + m_1(-q)]$, equal to $m_1(q)$ here by the inversion symmetry of the ring ($m_1(q) = m_1(-q)$), while for the Hermitian current J the two coincide unconditionally (App. A); only the kinetic term contributes, since the Hubbard interaction is diagonal in the occupation basis and commutes with ρ_q and S_q^z . On a Hubbard ring at $U/t = 8$ and half filling these operator moments reproduce the exact Lehmann moments of $S(q, \omega)$ and $S^{zz}(q, \omega)$ to $\sim 10^{-15}$ at every momentum, and the two forms of m_1 agree to the same precision. A reconstruction that injects a spurious low-frequency peak—in-gap for the gapped charge channel, near-zero for the gapless spin channel—while conserving the total weight passes the m_0 check (m_0 residual $\sim 10^{-15}$) yet fails the first moment by $\gtrsim 30\%$ once 40% of the band

weight is misplaced, firing the screen for both observables at every representative momentum. The moments themselves register spin-charge separation: the mean excitation frequency m_1/m_0 is set by the Hubbard scale U in the charge channel ($\approx 8.5t$ at $q = 2\pi/3$) and by the exchange $J = 4t^2/U$ in the spin channel ($\approx 1.0t$). These are exact-simulation demonstrations—the identical reconstruction-independent screen of Fig. 5, here applied to the collective channels of a half-filled $L = 6$ ring at $U/t = 8$ rather than to the doped current response—and their purpose is to demonstrate that the screen transports across observables, not to report a hardware result.

The same battery admits falsifiers of a different kind: any exactly known operator identity, evaluated directly on the prepared state, screens an error class the moment test does not reach. A clear example is the U(1) lattice-Schwinger Gauss law $\langle \sum_x G_x^2 \rangle = 0$, a native same-sample operator identity whose violation catches gauge-symmetry breaking—leakage out of the physical sector—orthogonally to the weight/subspace errors the

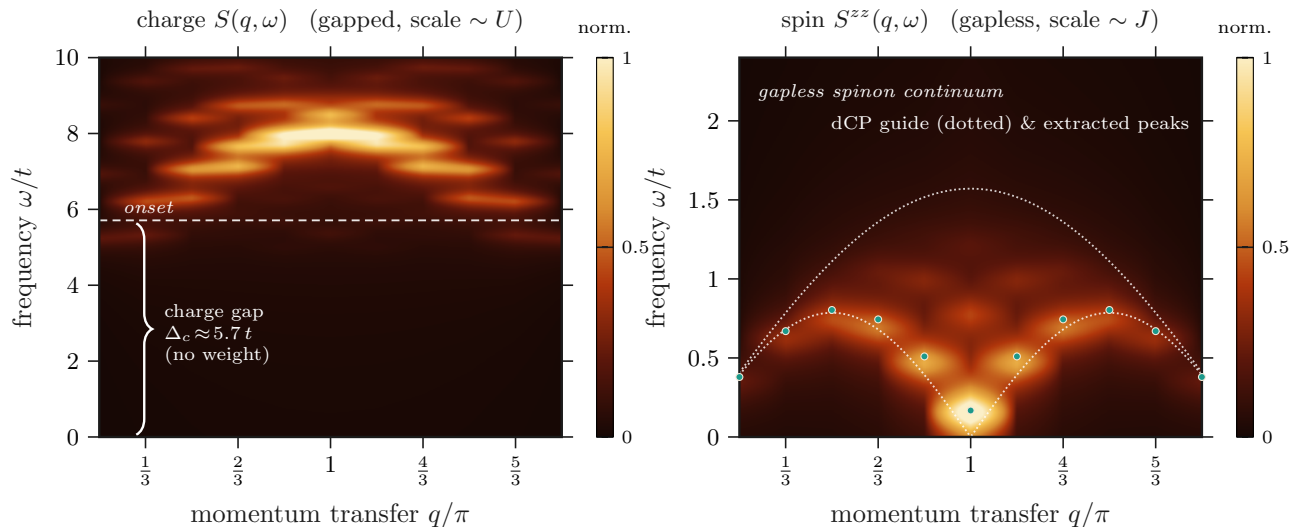


FIG. 4. **Spin-charge separation across the collective observables.** The charge dynamical structure factor $S(q, \omega)$ (left) is gapped by a particle-hole (density) excitation onset $\Delta_c \approx 5.7t$, a finite-size quantity distinct from the single-particle Mott gap $\mu^+ - \mu^- \approx 4.97t$ of Fig. 3 (the latter being the conventional many-body “charge gap”), both set by the Hubbard scale U and coinciding only in the thermodynamic limit—while the spin structure factor $S^{zz}(q, \omega)$ (right) carries gapless low-energy spinon weight, the hallmark of the gapless spin sector of the one-dimensional Mott insulator. The teal curves are the des Cloizeaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J/2)|\sin q|$ [91] and the upper Müller-continuum edge $\omega_U = \pi J|\sin(q/2)|$ [92], with $J = 4t^2/U = 0.5t$, and the open points are the dominant spinon peak *extracted* per momentum, which tracks the lower dCP edge—an ideal large- U Heisenberg *guide* over the eleven momenta $q/\pi \in \{1/6, \dots, 11/6\}$, not a strict bound (finite U , finite size and broadening $\eta = 0.12t$ push some weight below the ideal lower boundary; the field is linearly interpolated in q for display). Both are positive $T = 0$ Lehmann measures with their own moment sum rules, so the same reconstruction-independent screen applies to each [1, 2]. The two panels are drawn on different frequency windows— $\omega \lesssim U$ for the charge, $\omega \lesssim$ a few $J = 4t^2/U$ for the spin—because the two channels’ excitations live on parametrically separated energy scales; that separation is itself the visual signature of spin-charge separation, and it is why a single shared axis would render the spinon continuum illegible. Sector-basis Lanczos ground state with Haydock continued-fraction evaluation [74, 76], $L = 12$ sites at half filling (11 momentum transfers), $U/t = 8$; broadening $\eta = 0.20t$ (charge) and $0.12t$ (spin).

moment screen targets; because it is a state-level check rather than a screen on a reconstruction, we treat it as a complementary companion and develop it in Appendix D [34, 37].

IV. RESULTS II: INDEPENDENCE SUPPLIES THE DISCRIMINATING POWER

The central claim of the method is that *independence* is not a stylistic preference but the property that provides the discriminating power. We test it on the native failure mode of sample-based diagonalization: determinant-subspace truncation, in which the sampled subspace omits configurations that carry real spectral weight, so the reconstructed spectrum is internally consistent with an incomplete basis [14, 15]. We run two checks side by side on the same truncated data (Fig. 7). The independent estimator computes $\hat{m}_k = \langle O_k \rangle$ from an operator expectation that references the full operator algebra, not only the retained determinants; the back-computed control computes $\bar{m}_k$ by integrating the reconstructed $A(\omega)$, which lives entirely inside the truncated subspace. As truncation is tightened, the independent estimator’s dis-

crepancy Δ_k grows and crosses the falsification threshold, while the circular control’s discrepancy stays identically zero and reports the spectrum as consistent [15]. The circular control is blind by construction: it can only ever confirm that the reconstruction is consistent with itself. This is the concrete demonstration behind the Method’s warning, and it is the reason every moment in the screen must be estimated off the samples and never read back off $A(\omega)$ [1, 14]. One boundary must be stated precisely: the independent estimator applies the exact operator O_k to the *projected* state, so Δ_k detects spectral weight that leaks *outside* the retained determinant support; a corruption that redistributes weight strictly *within* the retained sector while preserving the moments used is, by construction, not caught by those moments. This residual is characterized and bounded rather than fundamental: on N distinct poles the weight-to-moment map is injective, so any within-sector redistribution preserving $m_0, \dots, m_K$ is refuted by some higher moment m_j ($K < j \leq N - 1$), and the deployed (m_0, m_1, m_2) battery already refutes any such redistribution large enough to move m_2 past its shot-budget threshold; which redistributions are detectable, and the shot cost of detecting them near a pole cluster, is settled by Theorem 2 (proved in Appendix E).

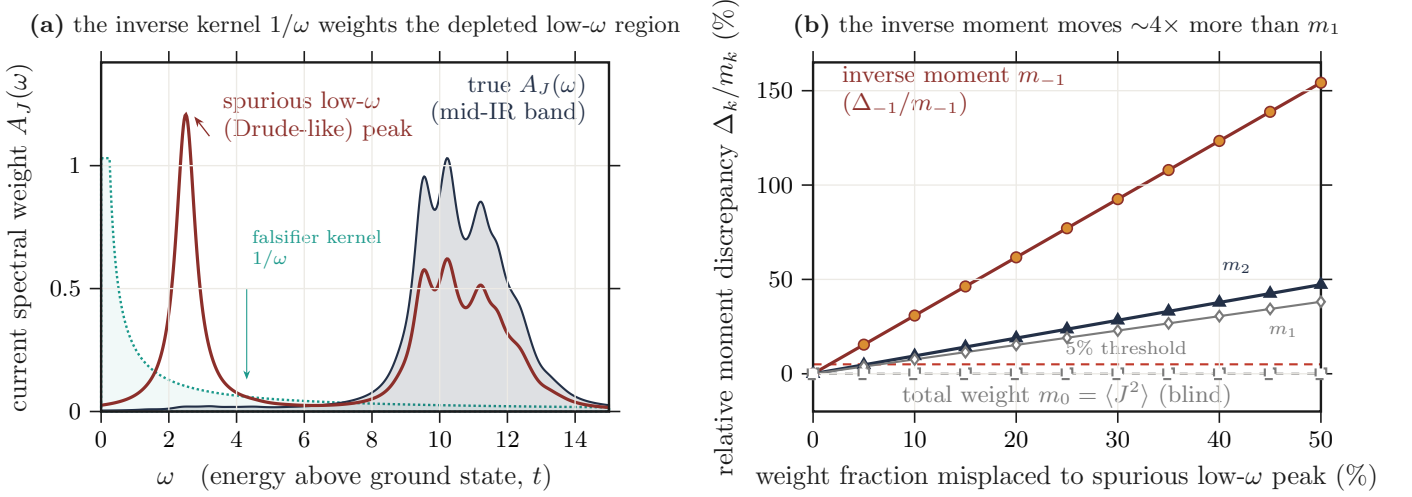


FIG. 5. **The current-response test: a wrong distribution of correct total weight, caught across moments.** A wrong reconstruction of the doped one-dimensional Hubbard current spectral function $A_J(\omega)$ ($L = 12$, $U/t = 8$, filling $2/3$, PBC; optical conductivity $\text{Re } \sigma \propto A_J(\omega)$) injects a spurious low-frequency (Drude-like) peak that transfers weight from the mid-infrared/Mott band while preserving the total $m_0 = \langle J^2 \rangle$. **(a)** The true regular spectrum (indigo) is a mid-infrared/Mott band with a nearly empty low- ω region (only 1.4% of m_0); the spurious peak (oxblood) is injected there, exactly where the inverse kernel $1/\omega$ (teal) weights most, and where the m_{-1} baseline ($m_{-1} = 0.324$) is smallest. **(b)** At each injected misplaced-weight fraction the total weight m_0 stays *blind* ($\Delta_0/m_0 < 10^{-13}$), so a total-weight check passes the wrong spectrum; the first moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$, estimated independently, *fires* (up to 38%), refuting it; and the inverse (f -sum) moment $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega = \frac{1}{2} \langle -\hat{T} \rangle - D$ (D the Kohn charge stiffness), whose $1/\omega$ kernel weights the depleted low- ω region, *fires* $\sim 4\times$ harder (up to 154%)—and is the cheapest moment to measure, $\langle -\hat{T} \rangle$ being one-body (verified against the exact Lehmann sum to $\sim 10^{-15}$). Sector-basis Lanczos ground state with Haydock/Ritz current poles [1, 32, 33, 74, 76].

The moment screen is thus a necessary-condition detector of support leakage and weight misassignment; the fine-tuned redistribution that preserves the entire deployed battery is the residual we disclose, not a witness we claim.

The independence test of Fig. 7 evaluates the reconstruction-independent estimator on the exact full-sector ground state, while only the reconstruction inhabits the truncated subspace—an idealization more independent than any device can be, since on hardware both branches are computed from one prepared state. We therefore repeat the test in its device-realizable form (Fig. 8): both moments are evaluated on the *same* subspace ground state $|0_S\rangle$ —the ground state of the truncated Hamiltonian $H_S = \Pi_S H \Pi_S$ —with the shared reference $e_0 = \langle 0_S | H | 0_S \rangle$. The independent estimator applies the *full* sparse operators, $\hat{m}_1 = \langle 0_S | J(H - e_0) J | 0_S \rangle$, whose action on $|0_S\rangle$ reaches determinants outside $\mathcal{S}$ (a classical post-processing of the shared state), while the reconstruction moment $\bar{m}_1 = \langle 0_S | J_S(H_S - e_0) J_S | 0_S \rangle = \int \omega A(\omega) d\omega$ stays within $\mathcal{S}$; their difference is precisely the operator leak of Eq. (B3) at $k = 1$. This shared-state leak $\Delta_1^{\text{shared}} = |\hat{m}_1 - \bar{m}_1|$ is a nonzero quantity that grows monotonically as the subspace is truncated and saturates near 100% at deep truncation: the qualitative screen still fires, driven by the real weight that $J|0_S\rangle$ leaks outside the retained support. The quantitative threshold is the device-realizable one: on the $L = 12$ doped ring

at $U/t = 8$ the shared-state discrepancy crosses the 5% falsification line at $d \approx 4.4 \times 10^4$ retained determinants (18% coverage), whereas the idealized full-sector-ground-state curve—which reproduces the crossing of Fig. 7 at $d \approx 1.6 \times 10^4$ (7%)—dips non-monotonically at moderate truncation where the full-state reference and the truncated reconstruction accidentally track one another. The device-realizable test is thus if anything the more conservative of the two at the operative threshold, and it removes the last idealization from the demonstration: the discrepancy arises from an operator leak a real device can measure, not from an unattainable reference state.

V. A PRE-REGISTERED TEST PROTOCOL

Every falsification shown so far targets an error we injected and knew in advance. To probe the screen where the outcome was not known to the classifier, we ran a blinded, pre-registered simulation. Before any instance existed we sealed—by SHA-256 hash—the battery (m_0, m_1, m_2 plus the interval-Hankel condition), the falsification thresholds (fixed from the `ibm.fez` shot budget of Appendix B), the error catalog, and the random seed; the generator and the classifier both re-verify the seal and refuse to run if it is broken, so no threshold could be tuned after seeing an outcome. The thresholds

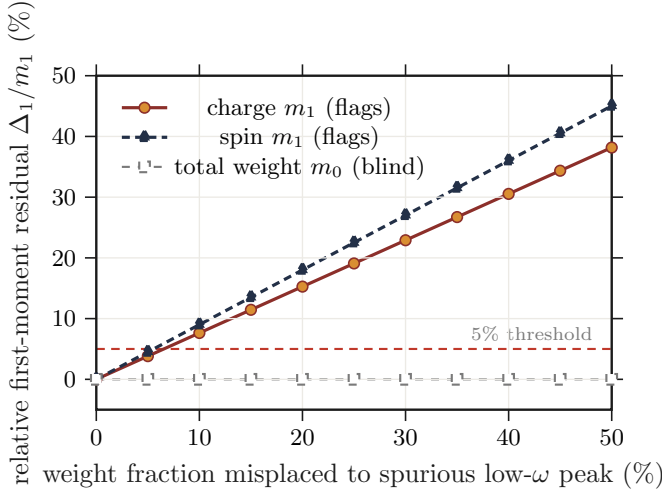


FIG. 6. **The moment test applied to the collective (charge and spin) channels.** The same reconstruction-independent screen of Fig. 5, applied without modification to the charge $S(q, \omega)$ and spin $S^{zz}(q, \omega)$ structure factors of Fig. 4 (whose spectra are shown there). For each channel a wrong reconstruction misplaces band weight into a spurious low-frequency phantom peak—in-gap ($\omega = 2.0t$) for the gapped charge channel, near-zero ($\omega = 0.10t$) for the gapless spin channel—conserving the total weight m_0 exactly. At $q/\pi = 2/3$ the total-weight sum rule m_0 (grey) stays blind for both channels, while the reconstruction-independent first moment $m_1 = \frac{1}{2}\langle[A^\dagger, [H, A]]\rangle$ rises through the falsification threshold for *both* the charge ($\rho_q = \sum_j e^{-iqj} n_j$, oxblood) and spin ($S_q^z = \sum_j e^{-iqj} S_j^z$, indigo) probes. The operator moments reproduce the exact Lehmann moments to $\sim 10^{-15}$ at every momentum $q/\pi \in \{1/3, 2/3, 1\}$. Hubbard ring $L = 6$, $U/t = 8$, PBC, half filling; exact simulation.

are fixed before the instances are generated and can be reused for other tests whose thresholds can be fixed in advance. The generator produced $G = 300$ instances drawn from a faithful control plus three *algorithmic* error causes—determinant-subspace (SQD) truncation, under-converged Krylov reconstruction, and an analytic-continuation artifact that injects a spurious spectral feature—and the classifier applied the frozen battery to each without access to its label.

The blinded outcome is informative precisely because it is uneven. Support-truncation errors were caught in every instance (56/56); spurious-feature artifacts in 74% (45/61) overall, which disaggregates into 93% (25/27) of the *total-weight-changing* artifacts—already reachable by the cheap diagonal m_0 channel—and 59% (20/34) of the *weight-preserving* artifacts, which only the off-diagonal moments can catch (the number that measures the off-diagonal channel’s added value); and the false-positive rate on faithful reconstructions was 0.9% (1/112, Wilson 95% CI [0.2, 4.9]%), well inside the nominal joint level. Under-converged Krylov reconstructions, by contrast, were missed at every truncation depth (0/71):

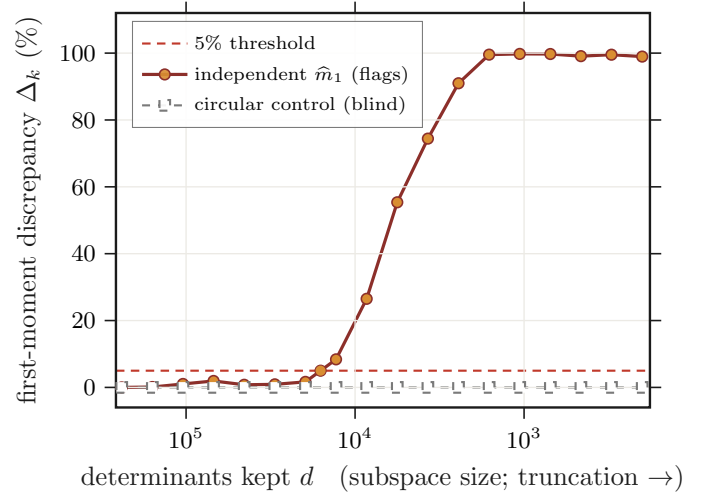


FIG. 7. **Independent vs. back-computed moment discrepancy under determinant-subspace truncation.**

The determinant subspace of the doped $L = 12$ Hubbard ring is truncated to its top d configurations by $|\psi_0|^2$ (the sample-based-diagonalization ordering; full support $n_{\text{sup}} = 2.4 \times 10^5$), and the current first moment is recomputed within it. On this truncated data the reconstruction-*independent* ground-state estimator’s discrepancy $\Delta_1^{\text{rel}} = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ stays negligible while the subspace is large but grows sharply once truncation becomes severe, crossing the falsification threshold at $d \approx 1.6 \times 10^4$ and saturating near 100%; a back-computed (circular) control read off the reconstructed spectrum stays identically zero and reports it consistent. The independent estimator therefore discriminates the subspace-truncation error class; the first moment stays *robust* until the subspace is small, so the screen catches severe truncation rather than mild undersampling. The circular control is blind by construction [1, 14, 15]. Sector-basis Lanczos: the truncated first moment is $\langle J\psi_0^d | (H_d - E_0^d) | J\psi_0^d \rangle$ from one subspace ground state, no full diagonalization [74].

a Lanczos representation reproduces the low-order moments *by construction*, so it passes a moment battery to within the assumed systematic bias. This refuted a secondary prediction we had sealed in the pre-registration—that the one-node Krylov case would be caught by m_2 —and the register was not revised. In this demonstration every rejection came from the moment discrepancies; the interval-Hankel positivity branch (a conservative point-PSD check on the interval endpoints) fired on none of the 300 instances, so the battery here reduces in practice to the moment test. The blinded test thus confirms, under conditions the classifier could not game, exactly the scope the method claims: it is a detector of support and structural errors and of spurious added features, not of reconstruction errors engineered or inherited to preserve the estimated moments. The scope is limited: the ground truth is classically known here and merely withheld from the classifier, so this is a blinded test, not a field catch of an error whose correctness is unknown to science, and

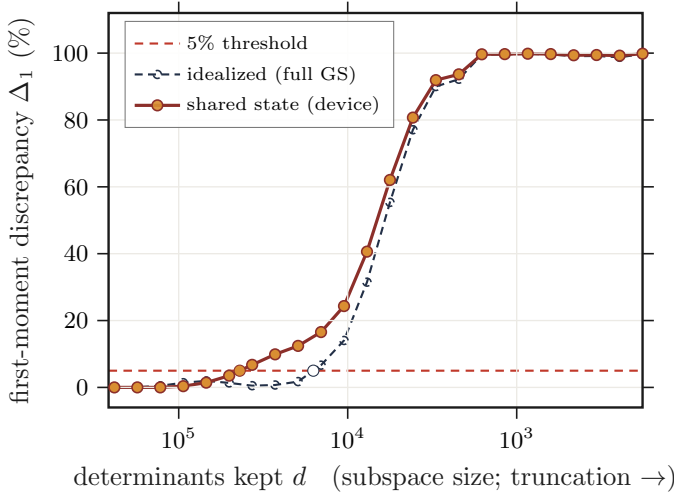


FIG. 8. **Device-realizable discrimination: the operator leak on a single prepared state.** Both first-moment branches are evaluated on the *same* subspace ground state $|0_S\rangle$ of the truncated Hamiltonian $H_S = \Pi_S H \Pi_S$, with shared reference $e_0 = \langle 0_S | H | 0_S \rangle$, exactly as a sample-based-diagonalization device would (Appendix B). Solid (amber): the device-realizable shared-state discrepancy $\Delta_1^{\text{shared}} = |\hat{m}_1 - \bar{m}_1|$, with $\hat{m}_1 = \langle 0_S | J(H - e_0) J | 0_S \rangle$ the independent estimator formed by applying the *full* operators to $|0_S\rangle$ (reaching determinants outside $\mathcal{S}$) and $\bar{m}_1 = \int \omega A(\omega) d\omega$ the within- $\mathcal{S}$ reconstruction moment—their difference is the $k=1$ operator leak of Eq. (B3). Dashed (indigo): the idealized curve of Fig. 7, in which $\hat{m}_1$ is instead evaluated on the full-sector exact ground state. As the retained determinant count d falls (truncation $\rightarrow$), the shared-state leak grows monotonically and crosses the 5% falsification threshold at $d \approx 4.4 \times 10^4$ (18% coverage), while the idealized curve crosses at $d \approx 1.6 \times 10^4$ (7%); both saturate near 100% at deep truncation, the signature of the operator leak. The idealized curve’s non-monotonic dip near $d \sim 4 \times 10^4$ is an accidental cancellation between the full-state reference and the truncated reconstruction, absent from the shared-state test. $L = 12$ doped Hubbard ring ($n_\uparrow = n_\downarrow = 4$, $2/3$ filling), $U/t = 8$, sector-basis Lanczos; the test rejects via the operator leak while the quantitative threshold is the shared-state one.

no device is in the loop—both remain future work.

These aggregates already separate the two channels by the errors they catch, and the reconstruction-independent moments carry that separation instance by instance (Fig. 9). Determinant truncation moves the total-weight sum rule $m_0 = \langle J^2 \rangle$, so the cheapest—and the only hardware-executable—channel catches it. The spurious-feature class is qualitatively different, and it splits in two. The *total-weight-changing* artifacts move m_0 and are already caught by the cheap diagonal channel (25/27). The *weight-preserving* subclass instead *preserves* the total weight and the determinant support by construction, so both diagonal channels are blind on it; there the off-diagonal moments carry the rejection (20/34 caught). Empirically, 24 sealed spurious rejections have

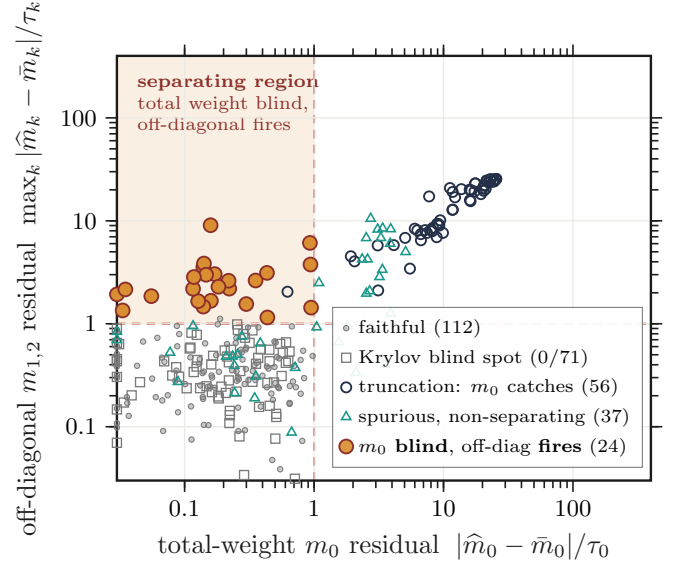


FIG. 9. **The diagonal and off-diagonal moments have complementary reach** (sealed, pre-registered blinded battery, SHA-256 0e4c3368...). For each of the 300 instances every residual $|\hat{m}_k - \bar{m}_k|$ is normalized to its frozen decision threshold τ_k (the pre-registered $z = 1.96$ rule at the `ibm.fez` shot budget, including the assumed 2% mitigation bias), so a residual > 1 is exactly a fire; the total-weight sum rule m_0 (the cheapest check, and the only channel executable on hardware) is on the horizontal axis, the larger of the two off-diagonal residuals m_1, m_2 on the vertical. Determinant-truncation errors (56/56 caught) sit at m_0 residual > 1 : the diagonal channel catches them (it also fires the off-diagonal, upper right). A weight-preserving subclass of the analytic-continuation *spurious-feature* artifacts keeps the total weight and the determinant support fixed by construction, so both diagonal channels are blind on it. On 24 sealed instances the m_0 residual falls below its threshold (shaded region) while an off-diagonal moment fires (m_1 on 15, m_2 alone on 9), carrying the rejection: the 20 weight-preserving artifacts (blind to m_0 by construction) together with 4 total-weight-changing ones whose weight shift stayed below the m_0 threshold (exemplar id 222: m_0 residual 0.16, m_1 residual 3.3, m_2 residual 9). Under-converged Krylov (0/71) preserves all low moments and is caught by neither—the disclosed blind spot; faithful reconstructions give one false positive (1/112). The thresholds here are shot-noise thresholds only: the off-diagonal moments are evaluated in noiseless simulation, and on current hardware they additionally carry the depolarizing bias of Sec. VI (~ 37 – 115% of the signal), so this is a structural separation, not device-realizable firing; their hardware evaluation is deferred.

the m_0 residual below its threshold yet an off-diagonal moment above its own (m_1 on 15, m_2 alone on the remaining 9): the 20 caught weight-preserving artifacts, together with 4 total-weight-changing ones whose weight shift fell below the m_0 threshold, so only the off-diagonal moments register them. This is the operational content of the completeness theorem (Theorem 2) and of Fig. 5

made concrete on blinded data: the off-diagonal moments *extend* the diagonal channel’s reach—truncation and total-weight errors are caught by both, while weight redistribution at fixed support and fixed total weight is caught only by the off-diagonal—so the off-diagonal estimator is not redundant machinery but the component that closes a corruption class the cheap channel cannot see. That component is, by the Sec. VI noise analysis, the one deferred to lower-error hardware; the diagonal coverage channel is what ran on `ibm_fez`.

VI. DEVICE-SIDE EXECUTABILITY AND COVERAGE-LEAK DEMONSTRATION ON IBM HERON

Beyond simulation, we evaluate the screen on real hardware. A first run establishes executability. Following the setup of Sec. II A, the sample-based-diagonalization spectral data of a companion study were measured on an IBM Heron processor (`ibm_fez`, $L = 6$, 12 qubits, 50,000 shots, TREX readout-error extinction + Pauli twirling + dynamical decoupling) [1]: the pipeline runs end to end and the sampled subspace recovers the *full* determinant sector ($|\mathcal{S}| = 300/300$), so the reconstruction coincides with exact diagonalization by coverage (Fig. 11). Precisely because the sector is fully covered, this run cannot exercise the falsifier—any moment of a full-coverage reconstruction equals the exact-state moment—and its raw counts were not retained; it establishes hardware executability and full-sector recovery only.

To exercise the falsifier on hardware we ran a second, purpose-built job that *retains* the raw per-shot counts. On an $L = 8$, 16-qubit electron-addition sector ($5 \uparrow, 4 \downarrow$; $U/t = 4$) we sample the same Krylov–Trotter flow ($K = 7$ times) as the companion at four shot budgets ($50/30/16/4 \times 10^3$ per circuit; TREX + twirling + DD) and post-select each pooled record to the physical particle/spin sector. The measured bitstrings define the support $\mathcal{S}$, and the reconstruction-independent coverage moment $\Delta_0 = |m_0 - \bar{m}_0| = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle$ is evaluated from the retained counts, with $m_0 = \langle cc^\dagger \rangle = 1 - \langle n_p \rangle = 0.5$ the exact one-sided addition-sector weight (agreeing with the Lehmann sum to 6×10^{-17}). As the budget falls and support coverage degrades from 94% to 35%, the residual grows, $\Delta_0 = 0.0044 \rightarrow 0.0323 \rightarrow 0.1232 \rightarrow 0.3726$ (Fig. 10): it is low for the near-complete reconstruction and high for the undersampled ones, evaluated from device counts against exact-known truth. This is a graded coverage residual, not a sealed-threshold pass/fire. The coverage probe’s own shot-noise floor is negligible ($\sigma \sim 10^{-4}$; point (d)), so every budget’s residual—including 0.0044 at 50k—lies orders of magnitude above it and would nominally register; but that floor is uninformatively tight for a support-completeness probe (it is set by counting statistics on $\langle n_p \rangle$, not by any accuracy scale), so the 50k point is best read as *low-residual*, below even the noiseless-sampling value 0.0238, rather than a formal

threshold pass (point (f)).

We are precise about what this establishes, in six points. (a) *It exercises the diagonal support-completeness channel only.* The sole device-derived quantity is the support set $\mathcal{S}$; the amplitudes entering $\bar{m}_0$ are the exact classical ones on $\mathcal{S}$, so Δ_0 gauges the completeness of the sampled support, not gate fidelity or amplitude accuracy—two devices reaching the same $\mathcal{S}$ with different frequencies return the same Δ_0 . (b) $\Delta_0 = m_0(1 - \bar{m}_0/m_0)$ is an identity, not a finding: it is the missed weight $\|(\mathbb{1} - \Pi_{\mathcal{S}})|\varphi\rangle\|^2$, the prefactor $\frac{1}{2}$ being m_0 of this orbital, not a universal law. (c) *The residual is comparable to a noiseless simulation of the identical circuits* (Fig. 10, dashed: $\Delta_0 = 0.0238/0.0451/0.1607/0.3191$). In this single run the device samples *more* distinct determinants than the noiseless circuit at the three higher budgets ($|\mathcal{S}| = 3682 > 3381$ at 50k), consistent with hardware noise broadening the sampled distribution so that post-selection retains a wider support; we report this as a single-run observation on the robust determinant count rather than a general claim, and note only that Δ_0 is by construction a support-coverage probe, not a gate-fidelity diagnostic. (d) *Shot noise is negligible* here (analytic $\sigma \sim 10^{-4}$); a nonparametric bootstrap of the retained counts gives an upward-biased upper envelope—inflated by the nonlinear support functional, not a confidence interval—reported with the reproduction materials. (e) *Support coverage and weight residual are distinct*: 94% is the configuration count $|\mathcal{S}|/|\mathcal{S}_{\text{sector}}| = 3682/3920$, while $\Delta_0 = 0.0044$ is a 0.9% weight residual (99.1% weight coverage). (f) *The 50k “clean pass” is a high-coverage, low-residual statement*— $\Delta_0 = 0.0044$, below the noiseless baseline 0.0238—not the crossing of an absolute shot-noise threshold, which for a coverage probe is uninformatively tight; the companion first moment Δ_1 (0.019/0.058/0.024/0.869) is *non-monotone* and shares the same hybrid status ($\bar{m}_1$ from the sampled support, reference m_1 exact-classical), so only Δ_0 is the monotone completeness probe.

This run removes the objection that the core instrument had never touched hardware—for the diagonal coverage channel—while leaving the discriminating step honest and explicit: the off-diagonal estimator (m_1, m_2 from rotated qubit-wise-commuting bond-basis settings, Fig. 12) is verified only in simulation (Appendix B). Its device-side evaluation is beyond present hardware for a rigorous reason, not merely engineering effort: the off-diagonal moment operator $O = c(H - E_0)c^\dagger$ is positive semidefinite, so its full-space average $\text{Tr}(O)/\dim \mathcal{H} \approx 6.8$ lies far above its ground-state value $m_1 \approx 2.1$ (a trace-to-value ratio ≈ 3.2), and depolarizing noise therefore biases the estimate *upward* toward that average. An exact density-matrix simulation of a representative bond-basis probe of the same trace-to-value ratio (a density probe with $m_1 = 10.6$, $\text{Tr}/\dim = 34.3$, ratio ≈ 3.2), under local two-qubit depolarizing noise at the `ibm_fez` rate ($\varepsilon \sim 2.5 \times 10^{-3}$), places this first-moment bias at $\sim 14\%$ of the signal at $\sim 10^2$ CZ, rising to $\sim 37\%$ at the companion

circuit depth (~ 290 CZ). This is exact for the surrogate and a *lower* bound on the true device floor: a static low-weight noise carrier omits the operator spreading of a real Krylov–Trotter circuit, which is expected to drive the bias toward the global-depolarizing value (~ 50 – 115% of the signal over the same depth range). Either bound is a substantial fraction of the signal—precluding a device-side catch of subtle reconstruction errors at this gate error—and the particle-number post-selection that protects the diagonal channel does not survive the bond-basis rotations. A device-side discriminant catch thus awaits a processor with lower gate error and is the next experiment; no device datum here bears on off-diagonal content. Table II summarizes what is demonstrated on hardware, what is simulation, and what is deferred.

The off-diagonal infeasibility rules out only the first moment $m_1(k)$, whose estimator is irreducibly off-diagonal; it does not close the door on a *discriminating* on-device channel, and it points to what such a channel must avoid. A diagonal discriminant exists: the momentum distribution $n_k = \int_{\omega < 0} A(k, \omega) d\omega = \langle c_{k\sigma}^\dagger c_{k\sigma} \rangle$, the occupied spectral weight at momentum k . Unlike the total spectral weight $\int A(k, \omega) d\omega = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$, pinned to unity at every k and hence blind to how weight splits between the occupied and empty sides, n_k carries, in the doped (metallic) electron-addition sector measured on device, a Fermi-surface crossover—a finite-size quasi-step of height $\max_k n_k - \min_k n_k \approx 0.8$ – 0.9 at $L = 6$, $U/t = 4$ – 8 (committed script `nk_falsifier.py`; in the thermodynamic limit the 1D step softens to a Luttinger power-law cusp)—so a reconstruction that flattens the dispersion moves $\hat{n}_k$ toward the flat filling value while the true n_k keeps the step. (At half filling the chain is a Mott insulator and n_k is analytic, so this discriminant targets the doped sector, not the half-filled spectrum of Fig. 3.) Crucially n_k is a *number* operator in the momentum basis, read out diagonally after a number-conserving fermionic Fourier transform (a Givens network, $L(L-1)/2$ rotations per spin sector), so the particle-number/spin post-selection that protects the diagonal coverage channel *survives* the basis change. The bond-basis rotations of m_1 , by contrast, carry penalties specific to the off-diagonal setting—the positive-semidefinite trace-to-value bias, operator spreading, and the loss of number/spin post-selection under those rotations—all of which this diagonal channel avoids, which is why its 37–115% bias floor has no counterpart here. Its *incremental* measurement cost is correspondingly small: an exact density-matrix estimate at the `ibm.fez` rate ($\varepsilon \sim 2.5 \times 10^{-3}$) places the flattening bias contributed by the ~ 60 -CZ transform itself at $\sim 9\%$ of the step (a connectivity-free lower bound, before fermionic-SWAP routing). We are explicit that this is not a defeat of the m_1 floor. $\hat{n}_k$ is not immune to noise—it flattens toward the same filling value a dispersion corruption does, so it discriminates only corruptions whose flattening *exceeds* the device’s own—and, like any device observable, it inherits the depolarization of the

TABLE II. **What is demonstrated on hardware, in simulation, analytically, and deferred.** The only device-derived quantity is the diagonal support set $\mathcal{S}$ (the coverage channel); every *discriminating* off-diagonal result is classically reproducible simulation, and the beyond-classical field catch is deferred. This separates the hardware content from the simulation content.

quantity	source	status
diagonal support set $\mathcal{S}$; coverage residual Δ_0 , $L = 8$ addition sector	<code>ibm.fez</code> ($\mathcal{S}$)	support measured on device; graded coverage residual, not a discriminating pass (Fig. 10)
full-sector recovery, $L = 6$	<code>ibm.fez</code>	executability precursor (Fig. 11)
off-diagonal moments m_1, m_2 (weight discrimination)	simulation	deferred to lower-error hardware (Sec. VI; Fig. 9)
blinded pre-registered false-positive / detection rates	simulation	calibrated (truth withheld from classifier)
Christoffel distance bound; necessary $\circ$ sufficient composition	miss- exact / analytic	bound proven; composition shown at the exact-moment level
field catch of an unknown-truth error (beyond-classical)	—	future work (Sec. IX)

deep state preparation, which at the companion Krylov depth is itself substantial and is common to the whole pipeline. What the diagonal channel changes is the *kind* of cost: it replaces the off-diagonal-specific penalties (operator spreading, post-selection loss) with the generic preparation-depth cost, so that a lower-depth preparation or a lower-error device turns n_k into a decisive on-device discriminant where m_1 cannot follow. This is a simulation forecast at the depolarizing-model level—readout error and coherent/leakage noise, which need not flatten toward filling, are not modeled, and the post-selection carries a sampling overhead—and n_k has not been run on hardware; its deployment, alongside the deferred m_1 catch, is the concrete next experiment.

VII. A CLOSED-FORM BOUND: THE CHRISTOFFEL MISS-DISTANCE

Having shown empirically what the screen catches, we now characterize analytically what a *passing* screen certifies: the screen carries a quantitative bound, drawn entirely from the classical theory of moments and orthogonal polynomials; the contribution is its deployment as a diagnostic for a quantum-computed spectrum, not the mathematics—the Christoffel–Darboux kernel is by now an established data-analysis tool in its own

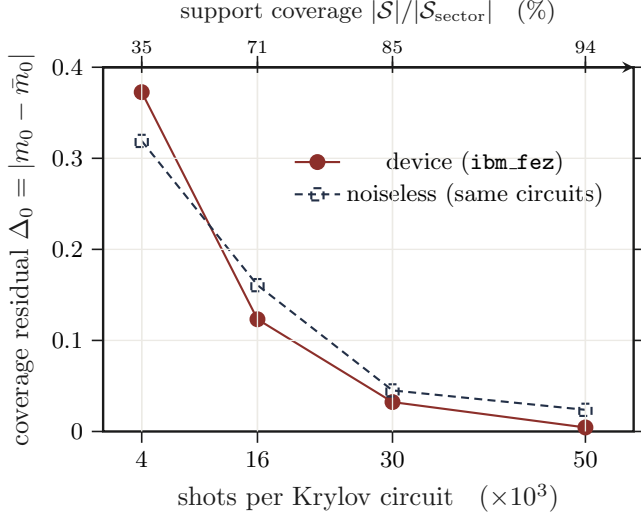


FIG. 10. **Device-side coverage-leak curve on IBM Heron** (`ibm_fez`, $L = 8$, 16 qubits, $U/t = 4$, $N+1$ addition sector, $K = 7$ Krylov–Trotter times, TREX + twirling + DD, retained per-shot counts). The reconstruction-independent coverage residual $\Delta_0 = |m_0 - \bar{m}_0|$ (solid, from device counts) grows as the shot budget—and hence support coverage (top axis, $|S|/|S_{\text{sector}}|$)—falls: it is low for the near-complete reconstruction (94% coverage, $\Delta_0 = 0.0044$) and high for the undersampled ones, evaluated against exact-known truth (a graded coverage residual, not a sealed-threshold pass/fire; see main text). A noiseless simulation of the identical circuits (dashed) sits above the device at the three higher budgets; in this single run the device samples more distinct determinants ($|S| = 3682 > 3381$ at 50k), consistent with noise broadening the support. By construction Δ_0 is a support-completeness probe, not a gate-fidelity diagnostic; shot noise is analytically negligible ($\sigma \sim 10^{-4}$). This purpose-built job (retained counts) is our own; it reuses the companion’s Krylov–Trotter circuits [1].

right [70, 71, 95, 96]. Consider the cumulative spectral weight $F(t) = \int_{-\infty}^t A(\omega) d\omega$ of a diagonal $T = 0$ Lehmann measure, and suppose the first $2n+1$ moments $m_0, \dots, m_{2n}$ are known exactly, which fix the orthonormal polynomials $p_0, \dots, p_n$ via the Hankel system [66]. Let $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ be the Christoffel function, also written $\lambda_n(t) \equiv W_n(t) = 1/K_n(t, t)$ with $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ the Christoffel–Darboux kernel; its explicit construction from the moments is given in Appendix C [70, 71].

Theorem 1 (Cumulative-weight miss-distance bound). *Let $d\mu = A(\omega) d\omega$ be a positive measure of compact support (a diagonal $T = 0$ Lehmann spectrum) whose Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^n$ is positive definite, and suppose the moments $m_0, \dots, m_{2n}$ are known exactly. Define the cumulative-weight miss-distance at t as the extremal spread $\sup\{|F_1(t) - F_2(t)|\}$ over all positive measures μ_1, μ_2 sharing those moments. Then for every t in the open interval $(\omega_{\min}, \omega_{\max})$ spanned by $\text{supp } \mu$ —the interior of its convex hull, equivalently the range of nodes*

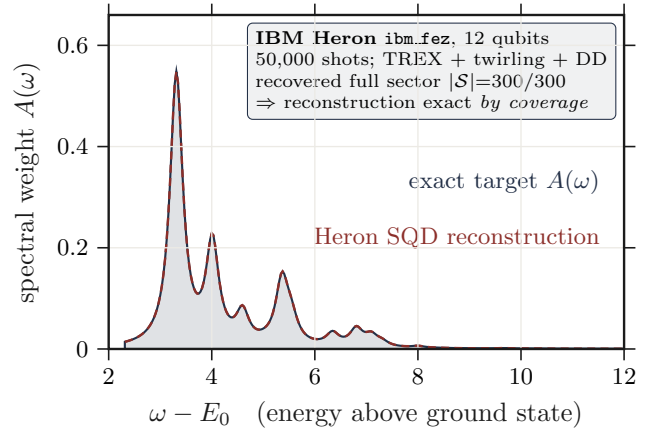


FIG. 11. **Executability precursor ($L = 6$, full-sector recovery)**. An earlier full-coverage run establishing only that the pipeline executes on hardware; the graded coverage test that actually exercises the falsifier from retained counts is Fig. 10. At the demonstrated size— $L = 6$ Hubbard, $U/t = 4$, $N+1$ electron-addition sector ($4 \uparrow, 3 \downarrow$)—a real IBM Heron run (`ibm_fez`, 12 qubits, 50,000 shots, TREX + Pauli twirling + dynamical decoupling) recovered the *full* determinant sector ($|S| = 300/300$), so the sample-based-diagonalization reconstruction coincides with exact diagonalization *by coverage* (it lies on the exact curve). The run establishes hardware executability and full-sector recovery, not a device-error line shape or a fidelity claim [1, 15].

at which both the upper and lower principal (Gauss–Radau) representations with a prescribed atom at t exist, so that interior atoms and the gap points between them are included—this extremal spread equals the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ —the maximal mass a moment-matching measure can place at t , i.e. the Gauss–Radau (Christoffel) weight $\lambda_n(t)$; and for any single reconstruction consistent with the moments the deviation from the true $F(t)$ is bounded above by $W_n(t)$ at every $t \in \mathbb{R}$ (proved in Appendix C) [97, 98]. The equality is stated on the convex-hull interior rather than on $\text{supp } \mu$ because the support of an atomic finite- N Lehmann spectrum has empty interior in $\mathbb{R}$; the everywhere-valid one-sided upper bound $|F_{\text{rec}} - F_{\text{true}}| \leq W_n$ that the screen actually deploys is unaffected by this distinction. The bound is monotone, $W_{n+1}(t) \leq W_n(t)$, since $K_{n+1} \geq K_n$. For a fixed finite Lehmann spectrum of N poles the construction is defined only for $n \leq N-1$; at a pole ω_i one then has $W_n(\omega_i) \downarrow w_i > 0$ (the residue), so W_n does not vanish on the support. Pointwise vanishing $W_n(t) \rightarrow 0$ is instead a thermodynamic-limit statement: if the discrete spectrum densifies to an absolutely continuous density A on a Stahl–Totik regular support S obeying the local Szegő condition $\int_I \log A > -\infty$, then $nW_n(x) \rightarrow A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [79], reducing on a single interval to Máté–Nevai–Totik [94].

The extremal ordinate spread at t equals the maximal atom a moment-matching measure can carry at t —

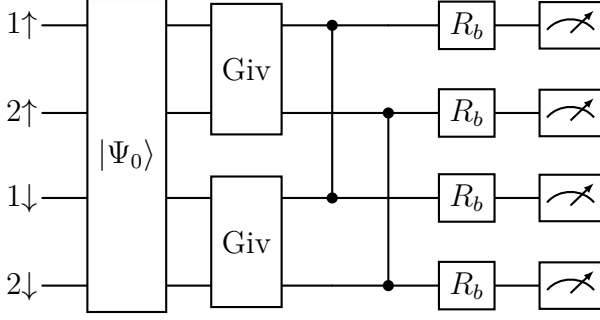


FIG. 12. **The device-side moment-estimation protocol.** One measurement circuit of the reconstruction-independent estimator, drawn as a two-site unit cell (4 qubits, two spin-up and two spin-down orbitals; the deployed run is $L = 6$, 12 qubits). A Trotterized state preparation from a reference state $|\Psi_0\rangle$ —nearest-neighbor Givens hopping ($\text{Giv} \equiv \text{hop } t$) within each spin sector and the on-site interaction ($R_{zz} = U \hat{n}_\uparrow \hat{n}_\downarrow$, the control-pairs coupling each up-site to its down-site)—is followed by a per-qubit bond-basis rotation R_b (a Hadamard-type rotation H or $S^\dagger H$ for the X/Y Pauli components, the identity for Z) and a computational-basis measurement whose per-shot counts are retained. One circuit is executed per qubit-wise-commuting Pauli group: the diagonal density probe yields m_0 from a single (all- Z) setting, and the off-diagonal content of m_1, m_2 from the remaining bond-basis settings (357 groups at $L = 6$; 83 suffice for the primary m_0, m_1 falsifier at $L = 12$). On real hardware only the state preparation and the sampler backend differ from the simulated verification of Appendix B.

the Gauss–Radau (Christoffel) weight $\lambda_n(t) = W_n(t)$ —which is the classical content of the Chebyshev–Markov–Stieltjes inequalities and Gauss–Christoffel moment theory, proved in Appendix C [66, 67, 70]. For t outside the convex hull the extremal-spread *equality* need not hold, but the one-sided bound the screen deploys, $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$, persists there as a direct consequence of the Markov–Stieltjes inequalities bounding the cumulative distribution of any moment-matching measure (App. C). This yields a computable resolution *bound*. Passing the moment battery through order $2n$ pins the cumulative weight $F(t)$ to within $W_n(t)$, a number we can evaluate, and enforcing more moments provably shrinks it [70, 79]. We use “bound”, not “certify”, deliberately: the screen never certifies that $A(\omega)$ is correct—consistent with its falsify-not-certify stance—only that, if it passes, its integrated weight is trapped within $W_n(t)$. The boundary of the claim is equally explicit: the bounded object is cumulative, integrated weight, not the pointwise value of $A(\omega)$ nor the position of any peak [70]; and the contraction *rate* is conditional on the regularity hypothesis and is not unconditional [79, 94]. The top panel of Fig. 13 makes the mechanism visible: the true cumulative weight $F(t)$ lies trapped inside the Gauss–Radau extremal envelopes over all measures sharing the moments, and the width of that bracket

is exactly $W_n(t)$, which narrows as n grows; its lower panels (a)–(c) show $W_n(t)$ itself contracting, its maximum shrinking monotonically, and its saturation at the dominant atom’s residue. We are explicit about the operational cost of this tightening: because $O_k = J \text{ad}_H^k(J)$ involves the $2n$ -th moment, the estimator variance grows steeply with order, so although the bound is monotone in principle, the moment orders a shot budget can reach do not tighten it to an operationally useful width: at transportable orders $n \leq 4$ the cumulative weight is bracketed only to within $W_4 \approx 0.875$ of the total (i.e. the bound is loose, consistent with Fig. 13). At those orders it constrains cumulative weight only coarsely, and we present it as such rather than as a sharp certificate. It is worth separating the two things the moments do, because they have opposite sensitivity to order. *Refutation*—a moment discrepancy $\Delta_k > \delta_k$, or a violated Hankel/Stieltjes positivity condition—is sharp at every order: a single infeasible moment vector rejects the reconstruction outright, with no dependence on the width W_n . What W_n governs is only the complementary, *positive* statement—the resolution to which a *passing* reconstruction’s integrated weight is bracketed. The coarseness at reachable orders therefore attaches to the corroborating bracket, not to the falsifying power of the same moments; a spectrum can be decisively refuted at $n \leq 4$ even where the bracket it would otherwise earn is loose. This locates the coarseness rather than removing it: the resolution of the positive certificate remains bounded by W_n , and tightening it still demands the higher, noisier moments (Fig. 13). The looseness of the *global* bracket does not, however, transfer uniformly across frequency. The (regular, $\omega > 0$) current measure studied here concentrates its weight in a single mid-infrared/Mott-band peak near $\omega \approx 1.4U$ (residue ≈ 0.78), with a nearly weight-free high-frequency tail above it. Evaluated at a frequency the measure leaves empty, $W_n(t^*)$ bounds the weight any moment-passing reconstruction can place there, and across that empty region the low-order width collapses— $W_1(t^*) \approx 2.4 \times 10^{-3}$, $W_2(t^*) \approx 2 \times 10^{-5}$, and $W_n(t^*) < 10^{-5}$ for $n \geq 3$ (App. C)—while the global maximum stays at $W_4 \approx 0.875$. The looseness of the global bracket therefore reflects the difficulty of resolving the dominant mid-band peak, not a coarse bound on out-of-band spurious weight, which the same shot-reachable moments constrain tightly. This is a computed property of this particular measure, not a theorem—a Gauss–Radau node may in general sit at an empty point and loosen the bound—so we established it by direct high-precision evaluation rather than by assumption, and we do not extrapolate it to the two-sided single-particle Mott gap, a distinct measure we did not test here.

Proposition 1 (Two-sided Markov–Krein window-mass bound). *Let $d\mu = A(\omega) d\omega$ be a positive measure (a diagonal $T = 0$ Lehmann spectrum) with $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$, and suppose $m_0, \dots, m_{2n}$ are known exactly. For any frequency window $[c, d]$ with c, d in the open convex-hull interior $(\omega_{\min}, \omega_{\max})$, the mass $\mu([c, d]) = \int_c^d A(\omega) d\omega$ of*

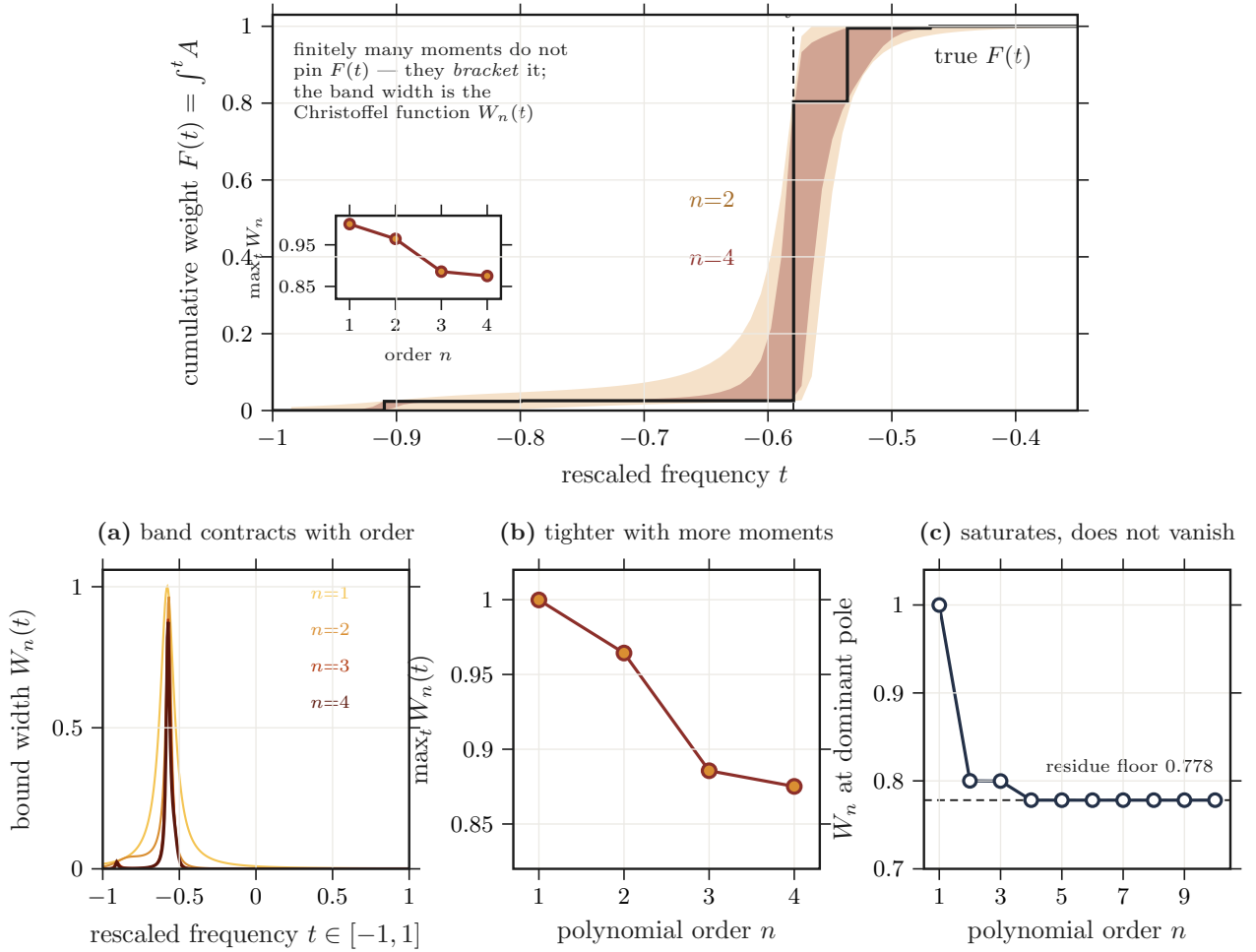


FIG. 13. **The Christoffel miss-distance bound.** *Top—the bracketing mechanism.* Finitely many moments do not pin the spectrum; they *bracket* its cumulative weight $F(t)$ (ink staircase) against the extremal Gauss–Radau measures sharing $m_0, \dots, m_{2n}$. The band width is the Christoffel function $W_n(t)$, and the $n=4$ band (deep) nests inside the $n=2$ band (light), so the certificate tightens with moment order. *Bottom (a–c)—the bound quantified.* (a) $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$ contracts as n grows; (b) its maximum over t shrinks monotonically $W_{n+1} \leq W_n$, from 0.9998 ($n=1$) to 0.875 ($n=4$); (c) at the dominant atom ($\omega = 11.09$, residue 0.778) W_n *saturates at the residue* and does not vanish—pointwise $W_n \rightarrow 0$ is a thermodynamic-limit statement for a densifying continuous density, not a property of a finite atomic spectrum. At these low, transportable moments the bracket is loose ($\max_t W_4 \approx 0.875$; a tight ~ 0.1 certificate needs ~ 20 – 30 moments); the bounded quantity is cumulative weight at interior points, not a line shape. Doped Hubbard $L = 6$, $U/t = 8$ current-operator Lehmann measure, rescaled to $t \in [-1, 1]$ [66, 67, 70, 71, 79, 94].

every positive measure sharing those moments obeys

$$\begin{aligned} \max\{0, F_-(d) - F_+(c)\} &\leq \mu([c, d]) \\ &\leq \min\{m_0, F_+(d) - F_-(c)\}, \end{aligned} \quad (1)$$

where $F_{\pm}(t)$ are the lower/upper principal (Gauss–Radau) cumulative representations of Lemma 3 (App. C): $F_-(t)$ is the mass the $(n+1)$ -point Gauss–Radau rule with a node fixed at t places strictly left of t , and $F_+(t) = F_-(t) + \lambda_n(t)$ with $\lambda_n(t) = W_n(t)$ its atom at t . The bracket is two-sided, and its width is at most $[F_+(d) - F_-(d)] + [F_+(c) - F_-(c)] = W_n(c) + W_n(d)$ (with equality before intersecting the trivial bounds $[0, m_0]$), so it inherits the monotone contraction $W_{n+1} \leq W_n$

of the Christoffel function and is tight wherever both endpoints fall in weight-poor frequency. The sharp Krein–Nudelman extremum—attained by a single principal representation with prescribed nodes at c and d , equivalently the exact-moment optimum of the certification semidefinite program of Refs. [41, 42]—is no wider, so (1) is a rigorous, computable two-sided certificate on integrated spectral weight, the two-sided strengthening of the one-sided point bound $|F_{\text{rec}} - F_{\text{true}}| \leq W_n$ the screen deploys (Chebyshev–Markov–Stieltjes inequalities; Krein–Nudelman, Karlin–Studden, Dette–Studden) [68, 70, 97, 98].

Evaluated on the same $L = 6$, $U/t = 8$ current Lehmann measure as the Christoffel bound (App. C;

committed script `markov_krein_window.py`), the two-sided bracket (1) confirms the one-sided picture from both sides. At the transportable maximum $n = 4$ ($m_0, \dots, m_8$) a window whose edge cuts the dominant mid-infrared peak inherits the full one-sided ambiguity— $\mu \in [0.070, 0.945]$, width $0.875 = \max_t W_4$ —whereas the empty high-frequency window (rescaled $[0.30, 0.90]$) is pinned to $\mu \in [0, <10^{-6}]$ and the band window flanking the peak (rescaled $[-0.75, -0.35]$, both edges in the gaps) to $\mu \in [0.9732, 0.9758]$ around the exact 0.9744. At the deployed order $n = 1$ (m_0, m_1, m_2) that out-of-band window already gives $\mu \in [0, 0.004]$, while an edge on the peak is uninformative ($\mu \in [0, 1]$). The honest reading matches the one-sided caveat: at shot-reachable orders the interval is as loose as $W_4 \approx 0.875$ only where a window edge crosses the dominant peak, and is tight ($\lesssim 10^{-3}$) for any window both of whose edges lie in weight-free frequency, so the two-sided bound sharpens the certificate exactly where the one-sided W_n is already tight and inherits its looseness at the peak. This window-mass bound is the real-axis, ground-state-moment counterpart of the Euclidean-time two-sided spectral bounds recently obtained via moment-problem semidefinite programs [63–65], differing in that our moments are sampled off a quantum device rather than read from lattice correlators. As with the one-sided bound we do not extrapolate this to the two-sided single-particle Mott gap, a distinct measure not tested here.

VIII. DETECTABILITY AND ITS SAMPLE-COMPLEXITY LIMIT

The Christoffel bound of Sec. VII governs the corroborating side of the screen—how tightly a *passing* spectrum’s integrated weight is bracketed. Its falsifying side admits an equally closed-form characterization: exactly which reconstruction errors the moment battery can and cannot detect, and, at a finite shot budget, how the power to detect them is set by sample complexity through the Vandermonde/Prony super-resolution limit. The limiting error class is the *within-sector redistribution*—a corruption that preserves the retained determinant support (so the leakage estimator of Sec. IV is blind by construction) and only reweights the reconstructed poles. The statement below is not a new mathematical theorem but an application: parts (a) and (b) repack the classical Prony/Vandermonde uniqueness of an atomic measure from its moments and the clustered-node super-resolution conditioning bounds [66, 99–102] as an explicit detectability-versus-shot-budget statement for the moment battery; we retain the **theorem** environment only for cross-referencing.

Theorem 2 (Detectability and completeness of the moment battery). *Let $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ be a diagonal $T = 0$ Lehmann measure on N distinct poles $\{\omega_n\}$ with $w_n = |(n|J|0)|^2 \geq 0$, and let a within-sector redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ ($\boldsymbol{\delta} \neq 0$, keeping $w_n + \delta_n \geq 0$) act*

on the same poles. Write $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ and induced shift $\Delta_k = |(V\boldsymbol{\delta})_k| = |\sum_n \delta_n \omega_n^k|$.

- (a) (Exact-arithmetic completeness.) *If the redistribution preserves $m_0, \dots, m_K$ with $K < N - 1$, then it is detected by some higher moment, $\Delta_j > 0$ for at least one j with $K < j \leq N - 1$; quantitatively*

$$\max_{K < j \leq N-1} \Delta_j \geq \frac{\sigma_{\min}(V_{0:N-1})}{\sqrt{N-1-K}} \|\boldsymbol{\delta}\|_2, \quad (2)$$

with $V_{0:N-1}$ the $N \times N$ node-Vandermonde on rows $0, \dots, N - 1$. Consequently the order- $(N-1)$ battery is complete for fixed-pole redistributions: no nonzero within-sector redistribution preserves all of $m_0, \dots, m_{N-1}$. If the poles are also allowed to move (a different N -atom measure), completeness holds at order $2N - 1$ by Prony parameter counting [66, 99]. The only alteration invisible to the entire moment sequence is weight traded among poles at a single (degenerate) frequency, which leaves $A(\omega)$ itself unchanged and is not an error.

- (b) (Operational detectability and its super-resolution threshold.) *At a shot budget giving per-moment noise floor $\delta_k = \sqrt{\text{Var}(O_k)/N_s + b_k^2}$ and two-sided threshold $\tau_k = z_{1-\alpha/2} \delta_k$, the redistribution is detectable at order j only if $\Delta_j > \tau_j$; the worst such shift over all $m_0, \dots, m_K$ -preserving redistributions is the linear program (E1), and detection at order $K+1$ holds iff $\Delta_{K+1}^{\max} > \tau_{K+1}$. In the sub-Rayleigh regime, where s poles cluster within minimal separation η (a pole-separation scale, distinct from the noise floors δ_k) small against the spectral scale, the relevant Vandermonde block obeys the super-resolution law $\sigma_{\min} \sim c_s \eta^{s-1}$ (up to a constant c_s fixed by s and the cluster position) [100–103]. Combined with (2), a within-cluster redistribution moving weight $\rho = \|\boldsymbol{\delta}\|_1$ whose first exposable order is $j = s - 1$ is detectable only if*

$$\begin{aligned} \rho &\gtrsim \frac{\tau_{s-1}}{c_s \eta^{s-1}}, \quad \text{equivalently} \\ N_s &\gtrsim \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{(c_s \rho \eta^{s-1})^2 - z_{1-\alpha/2}^2 b_{s-1}^2}, \end{aligned} \quad (3)$$

so the shot cost to falsify a within-cluster redistribution of fixed magnitude diverges as $\eta^{-2(s-1)}$ as the poles merge, at the super-resolution rate.

Proof sketch. Part (a): distinct poles make $V_{0:N-1}$ an invertible Vandermonde, so $V_{0:N-1}\boldsymbol{\delta} = 0$ forces $\boldsymbol{\delta} = 0$; preserving $m_0, \dots, m_K$ zeroes the first $K+1$ components of $V_{0:N-1}\boldsymbol{\delta}$, whence $\|(\Delta_{K+1}, \dots, \Delta_{N-1})\|_2 = \|V_{0:N-1}\boldsymbol{\delta}\|_2 \geq \sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2$, giving (2) after $\ell_2 \rightarrow \ell_\infty$; the moving-pole count is the classical Prony/Hankel uniqueness of an N -atom measure from $m_0, \dots, m_{2N-1}$. Part (b) substitutes the clustered-node scaling of $\sigma_{\min}$ into (2) and the detection condition $\Delta_j > \tau_j$ with $\tau_j = z_{\delta_j}$. Full details,

the linear program (E1), and the numerical instantiation are in Appendix E.

Corollary (detection power vs. separation and shot count). Under the Gaussian estimator validated in Fig. 14 (skewness -0.04 , excess kurtosis -0.05), the test fires with power $\geq 1 - \beta$ at significance α iff $\Delta_j \geq z_{\alpha,\beta} \delta_j$ with $z_{\alpha,\beta} = z_{1-\alpha/2} + z_{1-\beta}$. Substituting the super-resolution bound $\Delta_{s-1} \sim c_s \rho \eta^{s-1}$ yields $N_s \gtrsim z_{\alpha,\beta}^2 \text{Var}(O_{s-1}) / [(c_s \rho \eta^{s-1})^2 - z_{\alpha,\beta}^2 b_{s-1}^2]$, valid only above the bias-limited separation $\eta_* = (z_{\alpha,\beta} b_{s-1} / c_s \rho)^{1/(s-1)}$ below which no shot budget suffices (the systematic-bias floor). For the deployed $L = 6$, $U/t = 4$ battery the poles are well separated ($\eta_{\min} = 0.87$, $\sigma_{\min}(V_5) = 1.2 \times 10^{-2}$, no cluster), so conditioning is benign and detection is generous: a redistribution preserving m_0, m_1 moves m_2 by up to $\Delta_2^{\max} = 7.1$ against $\tau_2 = 1.848$ —a $3.8\times$ power margin (Table III). The residual bites only when the sampled sector carries a near-degenerate pair, where (3) makes the required N_s grow as $(\eta_{\min}/\eta)^2$ per unit reduction in separation. This is the sample-complexity face of the exact completeness of part (a): injectivity guarantees some m_j separates the redistribution, but the shot cost to realize that separation follows the Vandermonde super-resolution limit.

IX. DISCUSSION AND LIMITATIONS

The contribution of this work is a concrete method with characterized operating limits: a reconstruction-independent moment screen for quantum-computed spectral functions, a device-measurable estimator that realizes it at a real shot budget, a pre-registered protocol that calibrates its false-positive and detection rates where truth is known, and a closed-form resolution bound that fixes what a passing screen does and does not certify. Together these deliver the layer at which a quantum-computed spectrum can be shipped not as a certified answer but as a falsifiable hypothesis, accompanied by the tests it was given the chance to fail. What our screen delivers is corroboration in the strict sense—a spectrum that survived reconstruction-independent necessary-condition checks—and never proof of correctness. This is the correct reading of a necessary condition: passing raises credence and pinpoints the absence of a specific failure mode, while failing is logically conclusive [66]. The epistemic gain is real precisely because the estimator is independent of the reconstruction, so that agreement was not preordained [1].

The method must be positioned by mechanism against the two neighboring approaches, because superficially all three verify quantum results without a classical reference. Self-verifying variational simulation certifies a prepared *state* via an energy-variance bound measured on the device [37]; observable-estimation validation certifies error-mitigated *expectation values* by validating the hardware noise model and mitigation procedure without a classical

reference [26]. Our screen does neither: it certifies neither a prepared state nor an expectation value, and instead tests the internal consistency of a reconstructed spectral *function* against operator moments computed from the same samples, targeting the weight-misassignment failure mode of subspace methods that the other two approaches do not address [1, 15]. A distinct, non-physical layer verifies instead the numerical linear algebra of the classical diagonalizer that forms the reconstruction—cross-checking eigenpairs across independent LAPACK paths and sealing the result with tamper-evident certificates [104]—which catches software and floating-point error in the solver but, operating entirely on the classical computation, is by construction blind to the device-noise, sampling-incompleteness, and weight-misassignment errors that our reconstruction-independent screen is built to detect. The three physical-verification approaches are complementary—a hardware spectrum could, in principle, be produced under observable-estimation validation [26], prepared with self-verification of its input state [37], and then screened for weight misassignment by our moments—and we claim only the third role.

A referee may reasonably ask why we screen a finished reconstruction at all, rather than build the moment information *into* the reconstruction so that no inconsistent spectrum is produced in the first place. This is indeed the strategy of a mature family of analytic-continuation methods. Nevanlinna continuation imposes the exact interpolation and Pick/positivity structure of a causal response function, returning spectra that are consistent by construction [105]; maximum-entropy and stochastic analytic continuation regularize the ill-posed inversion subject to moment and sum-rule constraints supplied as prior information [61, 62]; and Prony, minimal-pole, and stochastic-pole-expansion methods fit the data to a compact pole model whose low-order moments can be pinned to their operator values [106, 107]. Where such a moment-constrained reconstruction is available it is preferable, and our screen does not compete with it. But the two are complementary by design. A moment-constrained reconstruction is consistent only with the moments *it* was given, only within the method that enforced them, and only for the imaginary-axis or time-domain data it was built from; a downstream consumer of a quantum-generated spectrum, by contrast, typically receives a finished $A(\omega)$ from a pipeline it does not control and cannot re-run. Our screen is built for exactly that setting: a reconstruction-independent necessary condition attachable to *any* spectrum, regardless of how—or whether—moments were used to produce it, drawing its discriminating power from moments sampled off the device rather than from the reconstruction. Building the moments into the reconstruction and screening the result against independently sampled moments are thus two layers of one defense, not alternatives.

A sharper form of this complementarity has recently appeared on the device side itself. Irmejs and Santos construct dynamical correlation functions directly on

hardware, without real-time evolution, from a continued-fraction representation whose level- k approximant is a weight- $O(k)$ operator—a spectral moment—measured on a prepared eigenstate, and prove exponential accuracy in a region of the complex-frequency plane bounded away from the real axis [108]; a related circuit-differentiation framework builds the retarded Green’s function from real-time-evolution circuit derivatives under realistic noise [109]. These are *constructive* programs: they use the same weight- $O(k)$ operator moments we exploit, but to *build* a trusted response function equipped with a priori convergence guarantees, provably accurate off the real axis. Our screen is their *diagnostic* dual. It assumes no control over how the spectrum was produced and issues no construction of its own; it takes a finished, real-axis $A(\omega)$ from a foreign sample-based-diagonalization or Krylov pipeline it cannot re-run, and asks only whether that function is consistent with independently sampled moments—precisely on the real axis, where the constructive guarantees are weakest. The two are complementary along the axis that matters: a constructive method earns trust where its convergence theorem reaches, whereas the diagnostic screen supplies a reconstruction-independent, on-axis necessary condition attachable to *any* spectrum, including those built by methods that carry no such guarantee [1].

The closed-form bound fixes exactly what the screen resolves. The miss-distance bound in cumulative spectral weight is the Christoffel function $W_n(t) = 1/\sum_{j=0}^n p_j(t)^2$, a classical object of the moment/orthogonal-polynomial theory, and it shrinks as the number of enforced moments grows [70, 71]. Its asymptotic contraction rate is governed by the $n\lambda_n$ behavior established by Totik for Christoffel functions of regular measures on the real line [79]—reducing on a single interval to Máté–Nevai–Totik [94]—which ties the resolution of our bound to well-understood approximation theory rather than to a heuristic. This is also the boundary of the claim: the bounded quantity is cumulative, integrated weight, not the pointwise value of $A(\omega)$ nor the position of any individual peak [70]. A screen that passes tells us the coarse-grained weight is placed consistently; it does not resolve a line shape [70, 75].

The limits are the following. First, necessary is not sufficient: infinitely many wrong spectra share any finite set of moments, so passing the screen can never certify correctness [66, 67]. Second, the moment screen addresses weight and subspace misassignment in a reconstruction, while the complementary Gauss-law falsifier addresses gauge-symmetry violation at the state level; neither is a general noise diagnostic, and device-noise trust remains the domain of observable-estimation validation [26]. Third, the positivity that underwrites the moment bounds is the positivity of a diagonal $T = 0$ Lehmann measure; off-diagonal or finite-temperature spectral functions are not positive measures in general and require separate treatment [35]. Fourth, our demonstrations use exact moments, an idealization; on hard-

ware, moments carry statistical and systematic error, and the correct object is a robust interval-moment test whose falsification threshold accounts for that uncertainty [26]. We regard the exact-moment results as the theory limit that the hardware-robust version must recover; Appendix B gives a sim-only Monte-Carlo demonstration of that robust version calibrated on the exact state, with the fully device-side deployment—propagating measured rather than assumed bias—left as the remaining step. A joint-sampling analysis of that noisy limit (reproduction materials) exposes a subtlety absent from exogenous-moment verification: because the independent estimator $\hat{m}_k$ and the reconstruction read-off $\bar{m}_k$ are computed from the *same* finite sample, the falsifier variance is $\text{Var}(\Delta_k) = \text{Var}(\hat{m}_k) + \text{Var}(\bar{m}_k) - 2\text{Cov}(\hat{m}_k, \bar{m}_k)$, with a same-sample correlation $\rho \approx 0.43$ at 84% coverage that decays to zero as coverage $\rightarrow 100\%$. Treating the moment as an independent input, as finite-statistics moment certificates do [41, 42], over-estimates this variance and renders the plug-in test silently over-conservative (realized false-positive rate $\sim 1.5\%$ at a nominal 5%) in the undersampled regime. We find, honestly, that this covariance is a *calibration* correction only: at matched false-positive rate the correlation-aware test yields no detection-power advantage over the independent-input treatment, so we report it as a caveat for same-sample verification pipelines, not a power gain. This same-sample structure is not new statistics but a quantum instance of Hausman-type specification testing [110]: $\hat{m}_k$ is the consistent, model-free estimator and $\bar{m}_k$ the model-implied one. The plug-in (independent-input) variance $\text{Var}(\hat{m}_k) + \text{Var}(\bar{m}_k)$ is exact only at full coverage—where the reconstruction is deterministic, $\rho \rightarrow 0$ —and is over-conservative when undersampled, where $\rho \approx 0.43$ drops a real -2Cov term; Hausman’s opposite, efficiency limit ($\text{Cov}(\hat{m}_k, \bar{m}_k) = \text{Var}(\bar{m}_k)$) instead collapses the falsifier variance to the difference $\text{Var}(\hat{m}_k) - \text{Var}(\bar{m}_k)$, and our correlation sits between the two. Reading the calibration as Hausman-type both explains the over-conservatism and locates it in a mature specification-testing literature rather than claiming it as new. Fifth, every error falsified here is synthetic—either an injected corruption or a controlled subspace truncation of otherwise-exact data. The blinded, pre-registered demonstration of Sec. V withholds the label from the classifier and so removes the tuning-after-the-fact concern, but the ground truth is still classically known; the screen has not yet caught an error in a spectrum whose correctness was unknown in advance, which is the field test we leave to future work. That same demonstration also delimits the reach of the screen: reconstruction errors that preserve the estimated moments—an under-converged Krylov representation being the clearest case, since it matches low moments by construction—pass the battery, consistent with the within-sector analysis of Appendix E.

A final objection concerns scale rather than method: the demonstrated systems—one-dimensional Hubbard rings at $L \leq 12$ —are precisely those that density-matrix

renormalization group (DMRG) and exact diagonalization solve outright, so no quantum resource is needed for their spectra. This is correct, and deliberate. These sizes are used because their exact moments are classically available, which is exactly what calibrating the screen’s operating characteristics requires: the screen is not in competition with DMRG or exact diagonalization here, it is characterized against them. Its intended regime is the one that motivates the quantum computation itself—real-time dynamics of two-dimensional and frustrated lattices, and *ab initio* molecular spectra—where no controlled classical spectral reference exists and a reconstruction-independent necessary condition is the only available cross-check. As evidence that the mechanism operates well beyond exact-diagonalization scale, the reproduction materials include a DMRG-oracle test at $L = 24$ (addition-sector dimension $\sim 10^{12}$) in which the coverage falsifier fires on subspace undersampling while a converged DMRG reference confirms the induced error; we present this only as a scalability check of the mechanism, not as a beyond-classical claim, since one-dimensional Hubbard remains DMRG-solvable. The beyond-classical field test—an error in a spectrum whose correctness is unknown to any classical method—is the experiment named above.

Several extensions follow directly. An interval-moment formulation, propagating shot noise and mitigation bias into a two-sided falsification band, is demonstrated on the exact state in Appendix B; carrying it to a full device-side deployment—with the mitigation bias measured and bounded from error-model violation rather than assumed [111]—is the immediate priority [26]. The Gauss-law check demonstrates that the framework accepts any exactly known operator identity as a native falsifier, which invites a catalog of domain-specific conditions—conservation laws, symmetry sectors, and sum rules—each screening a different error class [34, 75]. Extending positivity-based bounds to off-diagonal and finite-temperature spectra, where the naive Lehmann positivity fails, is the main theoretical opening [35]. Finally, because the moment machinery is exactly that of the kernel-polynomial, recursion, and quadrature methods [70, 75, 76], the screen composes naturally with the sample-based diagonalization and Krylov pipelines that generate the spectra it tests [1, 14–16, 112, 113]. The through-line is unchanged: a quantum spectrum earns trust not by being checked against an answer we do not have, but by surviving the tests it was given the chance to fail.

A. The verification stack: necessary $\circ$ sufficient

Because it needs only same-basis samples, the moment battery is near-zero-cost post-processing that composes with existing sample-based-diagonalization and Krylov workflows [1, 15]; its independent estimator branch is the reconstruction-free counterpart of predicting observables

from few measurements by classical shadows [50, 114], whose spectral specialization extracts gaps without ever forming $A(\omega)$ [115], and whose fermionic-adapted variant *constructs* dynamical correlation functions from at-most-two-copy shadows [116]—where our screen instead takes a finished $A(\omega)$ it did not produce and tests it, a diagnostic rather than a constructive use of the same samples. Formally, a reconstructed $T = 0$ Lehmann spectrum is a candidate solution to a truncated moment problem, so whether it may be trusted is decided by membership in a Hankel-positive convex body within the computable Christoffel miss-distance of Sec. VII, and thereafter corroborated, never proved [66, 67, 70]. The same non-commutative moment/positivity machinery has recently been turned into certified bounds on ground-state *observables* of many-body systems by Wang *et al.* [41], a property-certification approach adjacent to ours. The adjacency is sharpest when both are read not as separate programs on different objects but as two operations on *one* shared object—the ground-state moment sequence $\{m_k\}$, $m_k = \langle 0|O_k|0\rangle$, that our screen samples off the device and that the certification semidefinite program (SDP) bounds—for on that common sequence the composition is a well-defined mathematical statement on a single moment sequence. The program of Wang *et al.* solves a semidefinite relaxation of the ground-state problem to return *two-sided* intervals $m_k \in [\ell_k, u_k]$, rigorous upper and lower limits within which each moment must lie, from the same Hankel/localizing-matrix positivity we exploit [41], and Mortimer *et al.* carry it to finite measurement statistics, yielding such intervals valid under shot noise from informationally incomplete data [42]. That is a *sufficient* certificate: it brackets each m_k . Our moment screen acts on the identical $\{m_k\}$ from the opposite side, supplying only the *necessary* half: it estimates each m_k independently off the samples (Sec. II A) and fires whenever the estimated sequence is refutable—intrinsically, when its Hankel/Stieltjes geometry admits no positive measure (Fig. 2), or relative to the reconstruction, when the moments read off $A(\omega)$ miss the sampled or certified values. Three conditions thus bear on one moment sequence: SDP feasibility *brackets* each m_k ; our positivity geometry *rejects* any $\{m_k\}$ carried by no positive measure; and our reconstruction-independence *rejects* any $A(\omega)$ whose read-off moments fall outside that bracket. Feasibility certifies where infeasibility refutes, on the same coordinates—this is the precise content of necessary $\circ$ sufficient, and a satisfied screen still brackets nothing beyond the loose Christoffel width of Sec. VII. We make the composition concrete at the exact-moment level on the shared $\{m_0, m_1, m_2\}$ of the $L = 6$, $U/t = 8$ current measure: bare positivity bounds m_2 from below (refuting under-dispersion, $m_2 \geq m_1^2/m_0 = 123.3$) while the support-localizing certificate bounds it from above (refuting over-dispersion, $m_2 \leq (a+b)m_1 - abm_0 = 573.7$), so the two layers catch complementary corruptions of one sequence; running both on the same *device-estimated* $\{m_k\}$ is the nat-

ural next step.

The concrete object on which the two layers meet is the two-sided window-mass bound of Proposition 1. A moment-matrix relaxation certifying *integrated* spectral weight $\mu([c, d]) = \int_c^d A d\omega$ over a frequency window is exactly a two-sided interval on a linear functional of $\{m_k\}$: Proposition 1 evaluates that interval in closed form at exact moments (the Markov–Krein envelope, width $W_n(c) + W_n(d)$), and the SDP of Refs. [41, 42] would return its finite-statistics counterpart from the interval-valued m_k , against which our screen refutes any reconstruction whose window mass falls outside $[\mu_{\min}, \mu_{\max}]$. We do not attempt to lift the screen to a full SDP certification of frequency-resolved spectral *weights*, and we present this as a limitation rather than a stylistic choice: certifying a weight $|\langle n|J|0\rangle|^2$ at a resolved frequency bounds an *excited-state* matrix element, not a ground-state expectation, so it falls outside the ground-state moment hierarchy that makes the certification SDP tractable and would demand a separate positivity structure per excited sector. Certifying integrated window weight—the object both layers already share through $\{m_k\}$ —is therefore the natural bridge and a clear future direction: it upgrades the one-sided Christoffel envelope the screen currently deploys into a two-sided certificate, while the frequency-resolved line shape almost certainly remains beyond this machinery.

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Computations used exact diagonalization and matrix-product-state (TeNPy) tools; the $L = 6$ IBM Heron spectral reconstruction reprocessed here is that of the companion study [1], while the $L = 8$ coverage-leak job is our own.

DATA AVAILABILITY

The code and numerical data that support the findings of this study—the exact sector-Lanczos/Haydock engine, the interval-moment and joint-battery scripts, the committed circuit-scaling and estimator-verification scripts, the blinded pre-registration and its SHA-256 seal, the per-figure data, and a master notebook that regenerates every figure and number end to end—are openly available in the GitHub repository at <https://github.com/nicolasbonilla/reconstruction-independent-moment-tests>.

Appendix A: Derivation of the moment identities

We derive the operator form of the spectral moments used throughout, making explicit that each m_k is a ground-state expectation value computable without the dynamics. Let $H = H^\dagger$ have a non-degenerate ground

state $|0\rangle$, $H|0\rangle = E_0|0\rangle$, and a complete orthonormal eigenbasis $\{|n\rangle\}$ with $H|n\rangle = E_n|n\rangle$; let $J = J^\dagger$ be a Hermitian probe (a current or density). The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0)) \quad (\text{A1})$$

is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are $m_k = \int \omega^k A(\omega) d\omega = \sum_n (E_n - E_0)^k |\langle n|J|0\rangle|^2$.

Writing $|\langle n|J|0\rangle|^2 = \langle 0|J|n\rangle\langle n|J|0\rangle$, using $\langle n|(H - E_0)^k = (E_n - E_0)^k \langle n|$, and closing with completeness $\sum_n |n\rangle\langle n| = \mathbb{1}$,

$$m_k = \sum_n \langle 0|J|n\rangle (E_n - E_0)^k \langle n|J|0\rangle = \langle 0|J(H - E_0)^k J|0\rangle. \quad (\text{A2})$$

The ground state is annihilated by $H - E_0$, so acting to the right $(H - E_0)J|0\rangle = [H, J]|0\rangle$, and iterating,

$$(H - E_0)^k J|0\rangle = \text{ad}_H^k(J)|0\rangle, \quad \text{ad}_H(J) \equiv [H, J], \quad (\text{A3})$$

where ad_H^k denotes the k -fold nested commutator $[H, [H, \dots [H, J]]]$. Substituting Eq. (A3) into Eq. (A2) gives the closed operator form

$$m_k = \langle 0|J \text{ad}_H^k(J)|0\rangle \equiv \langle O_k \rangle, \quad (\text{A4})$$

with $O_k = J \text{ad}_H^k(J)$ built solely from H and J —no reference to the spectrum or to the reconstructed $A(\omega)$.

The lowest moments specialize immediately. For $k = 0$, Eq. (A4) gives the total weight $m_0 = \langle 0|J^2|0\rangle = \langle J^2 \rangle$. For $k = 1$, $m_1 = \langle 0|J[H, J]|0\rangle$; expanding the symmetric double commutator $[J, [H, J]] = 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J, [H, J]]|0\rangle = 2\langle 0|J(H - E_0)J|0\rangle = 2m_1, \quad (\text{A5})$$

so that

$$m_1 = \frac{1}{2} \langle 0|[J, [H, J]]|0\rangle. \quad (\text{A6})$$

This is the first-moment (double-commutator) sum rule of A_{JJ} ; for the current operator J it fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$, which is the m_{-1} moment of A_{JJ} (a ground-state kinetic energy) [2, 32, 33]. The symmetrized form (A6) is manifestly real and independent of the additive constant E_0 , which the one-sided expression $\langle 0|J[H, J]|0\rangle$ obscures.

Two features of Eq. (A4) make it the operational heart of the screen. First, O_k is an explicit operator polynomial in H and J ; expanded in the computational (occupation-number) basis, $\langle O_k \rangle$ becomes a weighted sum of basis observables whose ground-state expectation is estimated from the same sample set $\mathcal{S}$ that produced the reconstruction (Sec. II A)—this is the reconstruction-independent estimator $\hat{m}_k$. Second, the derivation used only (i) the positivity of the measure (A1) and (ii) the

ground-state annihilation $(H - E_0)|0\rangle = 0$; it therefore holds verbatim for any Hermitian probe and any Hamiltonian, which is the algebraic reason the screen transports across models.

For a non-Hermitian probe the same steps hold with $J^\dagger$ on the bra side, $m_k = \langle 0|J^\dagger(H - E_0)^k J|0\rangle$, so that $m_0 = \langle J^\dagger J \rangle$. The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 3 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$, and its moments close under *anticommutators* rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma} d\omega &= \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1, \\ \int \omega A_{k\sigma} d\omega &= \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A7})$$

the Hartree-shifted band centroid [2]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A7) is the version used for the single-particle screen of Fig. 3, and is likewise a ground-state expectation of explicit operators estimable from the same samples.

Derivation of the anticommutator moment identities. Equation (A7) is not a fresh sum rule but the non-Hermitian specialization of Eqs. (A2)–(A4), derived here for completeness. Write $d \equiv c_{k\sigma}$, $d^\dagger \equiv c_{k\sigma}^\dagger$, with $\{|n\rangle\}$ the complete eigenbasis of H over all particle-number sectors, $H|n\rangle = E_n|n\rangle$, ground state $|0\rangle$, energy E_0 . The two-sided single-particle spectral function is the sum of an electron-addition ($N+1$) and an electron-removal ($N-1$) branch,

$$\begin{aligned} A_{k\sigma}(\omega) &= \sum_n |\langle n|d^\dagger|0\rangle|^2 \delta(\omega - (E_n - E_0)) \\ &\quad + \sum_n |\langle n|d|0\rangle|^2 \delta(\omega + (E_n - E_0)), \end{aligned} \quad (\text{A8})$$

a non-negative measure whose support *straddles* $\omega = 0$ (addition poles at $\omega = E_n - E_0 > 0$, removal poles at $\omega = -(E_n - E_0) < 0$). Its power moments split accordingly,

$$\begin{aligned} \mu_p \equiv \int \omega^p A_{k\sigma} d\omega &= \sum_n (E_n - E_0)^p |\langle n|d^\dagger|0\rangle|^2 \\ &\quad + (-1)^p \sum_n (E_n - E_0)^p |\langle n|d|0\rangle|^2. \end{aligned} \quad (\text{A9})$$

Applying to each branch the steps of Eqs. (A2)–(A4)— $(E_n - E_0)^p \langle n|d^\dagger|0\rangle = \langle n|\text{ad}_H^p(d^\dagger)|0\rangle$ and completeness—gives the closed operator form

$$\mu_p = \langle 0|d \text{ad}_H^p(d^\dagger)|0\rangle + (-1)^p \langle 0|d^\dagger \text{ad}_H^p(d)|0\rangle, \quad (\text{A10})$$

built solely from H , d , and $d^\dagger$. For $p = 0$ this is the canonical anticommutator $\mu_0 = \langle 0|dd^\dagger + d^\dagger d|0\rangle = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1$. For $p = 1$, using $H|0\rangle = E_0|0\rangle$ and $\langle 0|H = E_0\langle 0|$ on the outer factors one verifies term by term that

$$\mu_1 = \langle 0|d[H, d^\dagger]|0\rangle - \langle 0|d^\dagger[H, d]|0\rangle = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle, \quad (\text{A11})$$

the anticommutator first-moment identity, manifestly real and independent of the additive constant E_0 . For the Hubbard Hamiltonian $H = \sum_{k'\sigma'} \epsilon_{k'} c_{k'\sigma'}^\dagger c_{k'\sigma'} + U \sum_i n_{i\uparrow} n_{i\downarrow}$ the commutator evaluates in closed form: the kinetic term gives $[c_{k\sigma}, H_0] = \epsilon_k c_{k\sigma}$, while $[c_{k\sigma}, H_U] = \frac{U}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma} n_{j,-\sigma}$ (with $c_{k\sigma} = \frac{1}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma}$), and since $\{c_{j\sigma} n_{j,-\sigma}, c_{j'\sigma}^\dagger\} = \delta_{jj'} n_{j,-\sigma}$,

$$\begin{aligned} \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle &= \epsilon_k \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle + \frac{U}{L} \sum_j \langle n_{j,-\sigma} \rangle \\ &= \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A12})$$

the Hartree-shifted band centroid quoted in Eq. (A7) and evaluated per momentum in Fig. 3 [2, 60].

Scope: which falsifiers transport to the single-particle channel. The zeroth- and first-moment checks carry over intact— μ_0 and μ_1 are genuine ground-state expectation values of explicit operators [Eqs. (A10)–(A11)], estimable from the same samples as reconstruction-independent falsifiers Δ_0, Δ_1 at each k —and so does the two-sided Hamburger–Hankel positivity $H_n = [\mu_{i+j}] \succeq 0$, since a positive measure on $\mathbb{R}$ is a Hamburger moment sequence irrespective of the sign of its support. The one-sided positivity falsifiers do *not* transport. Because $A_{k\sigma}$ is supported on both signs of ω (addition plus removal), the shifted-Stieltjes condition $H_n^{(1)} = [\mu_{i+j+1}] \succeq 0$ is unavailable—it presupposes support on $[0, \infty)$ —and the Christoffel/Gauss–Radau cumulative-weight certificate of Sec. VII, constructed on the diagonal one-sided ($\omega \geq 0$) Lehmann measure, is neither deployed nor extrapolated to this two-sided measure, consistent with the boundary flagged at the close of Sec. VII. In the single-particle channel the screen therefore reduces to the moment-discrepancy falsifiers Δ_0, Δ_1 and Hamburger–Hankel positivity; the Stieltjes-shift and Christoffel one-sided certificates remain the province of the diagonal current/density response.

Appendix B: The reconstruction-independent moment estimator

We give the explicit estimator $\hat{m}_k$ and show precisely why it is independent of the reconstruction and why it—unlike the spectral moment $\bar{m}_k$ —detects subspace truncation. Sample-based diagonalization returns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and, from the classically diagonalized projected Hamiltonian $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$ with $\Pi_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |x\rangle\langle x|$, the ground state $|0\rangle = \sum_{x \in \mathcal{S}} c_x |x\rangle$ and eigenpairs $\{E_n, |n\rangle\}_{n \in \mathcal{S}}$ [14, 15]. The probe J and the Hamiltonian H are sparse in this basis, with known matrix elements $J_{yx} = \langle y|J|x\rangle$ and $H_{yx} = \langle y|H|x\rangle$.

The moment $m_k = \langle 0|J(H - E_0)^k J|0\rangle$ of Appendix A admits two evaluations that coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace*

resolution $\Pi_{\mathcal{S}}$ in place of the identity yields the reconstruction (spectral) moment,

$$\begin{aligned}\bar{m}_k &= \sum_{n \in \mathcal{S}} (E_n - E_0)^k |\langle n | J | 0 \rangle|^2 \\ &= \langle 0 | J [\Pi_{\mathcal{S}}(H - E_0)\Pi_{\mathcal{S}}]^k J | 0 \rangle \\ &= \langle 0 | J (H_{\mathcal{S}} - E_0)^k J | 0 \rangle,\end{aligned}\quad (\text{B1})$$

which is exactly $\int \omega^k A(\omega) d\omega$ and lives entirely inside $\mathcal{S}$; the $k = 0$ term carries the subspace-algebra convention $[\Pi_{\mathcal{S}}(H - E_0)\Pi_{\mathcal{S}}]^0 \equiv \Pi_{\mathcal{S}}$ (equivalently $(H_{\mathcal{S}} - E_0)^0 \equiv \mathbb{1}_{\mathcal{S}} = \Pi_{\mathcal{S}}$), so that $\bar{m}_0 = \langle 0 | J \Pi_{\mathcal{S}} J | 0 \rangle$ is the coverage-truncated weight rather than the full $\langle J^2 \rangle$, consistent with the first line of Eq. (B1). The reconstruction-independent estimator instead applies the sparse operator $O_k = J(H - E_0)^k J$ directly,

$$\begin{aligned}\hat{m}_k &= \langle 0 | J (H - E_0)^k J | 0 \rangle \\ &= \sum_{x, x' \in \mathcal{S}} c_x^* c_{x'} \langle x | J (H - E_0)^k J | x' \rangle,\end{aligned}\quad (\text{B2})$$

where each matrix element $\langle x | O_k | x' \rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\hat{m}_k$ probes the action of J and H on $|0\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it.

The two agree exactly when the states generated from $|0\rangle$ by these operators— $J|0\rangle$, $(H - E_0)J|0\rangle$, and so on up to order k —already lie in $\text{span } \mathcal{S}$, so that $(\mathbb{1} - \Pi_{\mathcal{S}})$ annihilates them and $\Pi_{\mathcal{S}}$ may be dropped from Eq. (B1). Under determinant-subspace truncation this fails: writing $|\varphi\rangle = J|0\rangle$,

$$\hat{m}_k - \bar{m}_k = \langle \varphi | (H - E_0)^k | \varphi \rangle - \langle \varphi | [\Pi_{\mathcal{S}}(H - E_0)\Pi_{\mathcal{S}}]^k | \varphi \rangle, \quad (\text{B3})$$

the spectral weight the truncation leaks outside $\mathcal{S}$. At $k = 0$, under the same convention, this reduces to $\Delta_0 = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle = \langle \varphi | \varphi \rangle - \bar{m}_0 \geq 0$, the omitted weight itself—the coverage leak measured directly from retained device counts in Sec. VI. The discrepancy $\Delta_k = |\hat{m}_k - \bar{m}_k|$ is thus nonzero precisely to the extent that $\mathcal{S}$ omits configurations carrying weight: the independent estimator fires, while the circular control compares $\bar{m}_k$ to itself and returns $\Delta_k \equiv 0$, staying blind (Sec. IV).

Equation (B2) is equivalently an importance-weighted average over $\mathcal{S}$,

$$\hat{m}_k = \sum_{x \in \mathcal{S}} |c_x|^2 O_k^{\text{loc}}(x), \quad O_k^{\text{loc}}(x) = \frac{\langle x | O_k | 0 \rangle}{\langle x | 0 \rangle}, \quad (\text{B4})$$

the local-operator estimator familiar from variational Monte Carlo [117]; the local moment $O_k^{\text{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside $\mathcal{S}$. Because $\mathcal{S}$ and the amplitudes c_x are fixed by a finite number of measured shots and by the error-mitigation pipeline, $\hat{m}_k$ carries statistical and systematic uncertainty δ_k , and the exact-moment falsifier $\Delta_k > 0$ is

replaced on hardware by the robust interval-moment test $\Delta_k > \delta_k$ anticipated in the main text [26].

Here $\hat{m}_k$ evaluates a physical constraint using operator support that the reconstruction—built from the samples on $\mathcal{S}$ —may have omitted, so a spectrum that has misplaced weight cannot satisfy it unless the omitted physics was in fact absent.

The exact-moment falsifier acquires its hardware form here. Modeling the estimator uncertainty as $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$ (shot variance over N_s samples plus a mitigation-bias budget b_k), the two-sided decision is to falsify when $|\hat{m}_k - \bar{m}_k| > z_{1-\alpha/2} \delta_k$ at level α . The battery-level version is a semidefinite feasibility test: the spectrum survives if there exists a moment vector $\mu_\ell \in [\hat{m}_\ell - \delta_\ell, \hat{m}_\ell + \delta_\ell]$ whose Hankel matrix is positive semidefinite, $H_n \succeq 0$ (and, for the one-sided $T = 0$ support, the shifted $H_n^{(1)} \succeq 0$); infeasibility falsifies. This is the robust interval-moment test anticipated in the main text [26]; the per-demonstration operators, targets, and intervals accompany the reproduction materials.

The estimator also admits a directly device-native evaluation that we verify explicitly, since it settles how the off-diagonal (hopping) content of $\hat{m}_1, \hat{m}_2$ is obtained on hardware. For a probe diagonal in the computational basis—the density/structure-factor probe $\rho_q = \sum_j \cos(qj) n_j$ —the moment operator $O_k = \delta \rho_q (H - E_0)^k \delta \rho_q$ is a sum of Pauli strings, and $\hat{m}_k = \langle 0 | O_k | 0 \rangle = \sum_P c_P \langle P \rangle$ is a linear combination of Pauli expectations estimated from counts: m_0 is a pure computational-basis (counts-only) estimator, while the hopping terms in m_1, m_2 are read from a small number of qubit-wise-commuting “bond-basis” measurements (one rotated-basis circuit per commuting group), exactly as in shadow/Pauli-grouped observable estimation [26, 50]. We verify this estimator by finite-shot sampling of the exact ground state at the $N_s = 5 \times 10^4$ budget. On the $L = 6$ Hubbard system (12 qubits, matching the hardware demonstration) the counts-based $(\hat{m}_0, \hat{m}_1, \hat{m}_2) = (2.730, 10.55, 47.01)$ recover the exact $(2.737, 10.61, 47.64)$ within 0.7–0.9 standard errors, using 357 qubit-wise-commuting measurement bases; the smaller $L = 4$ system (8 qubits) reproduces the full battery within shot noise as well (every moment within one standard error of exact). The number of measurement bases grows polynomially with system size, being bounded by the locality of O_k rather than by the Hilbert-space dimension, so the estimator remains hardware-deployable at sizes where a classical statevector cross-check does not. The base count depends only on the operator’s Pauli support, not on the numerical moment values—the probe shift $\langle \rho_q \rangle$ vanishes by half-filling symmetry, so it can be enumerated without the exponential statevector; a committed script does so and reproduces the full-eigendecomposition count exactly where the latter is affordable ($L \leq 8$). The full (m_0, m_1, m_2) battery then needs 126, 357, 721, 1191, 1837 commuting bases at $L = 4, 6, 8, 10, 12$ (8 to 24 qubits), while the primary first-moment falsifier (m_0, m_1) —

the reconstruction-independent screen itself—needs only 19, 35, 51, 67, 83 bases, growing roughly linearly and far more slowly than the full battery. The measurement-basis overhead thus stays modest—83 settings for the first-moment falsifier at 24 qubits—rather than exponential. We are careful not to overstate this into a deployability claim: the counts-based moment recovery was validated only up to 12 qubits and the basis-count cross-check against full diagonalization only up to $L = 8$; the shot budget and device-side accuracy at 24 qubits remain to be characterized, and the device would carry the state preparation that the classical statevector cross-check cannot. The reconstruction-independent estimator, including its off-diagonal content, is thus device-measurable from a fixed shot budget *in principle*; on present `ibm_fez` hardware the off-diagonal m_1, m_2 estimate carries a $>40\%$ upward gate-error bias and awaits a processor with $\sim 10\times$ lower gate error (Sec. VI), so this shot-budget statement is not a claim of off-diagonal recovery on current hardware; on hardware only the state preparation and the sampler backend change, and the runnable job specifying the prep, the retained per-basis counts, and this estimator accompanies the reproduction materials.

We close with a direct Monte-Carlo demonstration of this test at the real `ibm_fez` scale and budget ($N_s = 5 \times 10^4$), computed sim-only by drawing $M = 4 \times 10^3$ replicas of N_s shots from the exact-state distribution $|\langle x|0\rangle|^2$ —not a device measurement, because the device bitstring counts of the companion run were not retained (Fig. 14). The demonstration *validates* the analytic interval rather than replacing it: the realized independent estimator is Gaussian (skewness -0.04 , excess kurtosis -0.05), its spread matches the analytic shot term to 1% (0.0799 vs 0.0806), and its realized false-positive rate at full reconstruction is 4.5% (Wilson-95% $[3.9, 5.2]\%$) against the nominal 5%; against synthetic truncated reconstructions the test reaches full detection power for coverage below $\sim 90\%$, with a single exception near 53% coverage where an accidental first-moment match lowers power, closed by the (m_0, m_1, m_2) +Hankel–Stieltjes battery. Two inputs are modeled, both in the safe direction. The shot term uses the exact-state local-estimator variance $\text{Var}(O_k) = \langle O_k^2 \rangle - \langle O_k \rangle^2$, a *lower bound* on the true per-shot uncertainty (which additionally carries ratio-estimator and Pauli-grouping overhead); the bias term b_k is a residual we *assume and sweep* rather than measure. For the doped $L = 6$, $U/t = 4$ current probe, $\text{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\text{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); combined with a 2% bias budget the 95% two-sided falsification threshold is $z_{0.975} \delta_1 \approx 0.23$ (5.4% of m_1). A principled, budget-informed threshold thus falls close to the 5% figure a practitioner would choose heuristically, corroborating that choice rather than replacing it. Within this threshold the forecast corroborates the converged reconstruction ($|\hat{m}_1 - \bar{m}_1| \sim 10^{-15}$) and refutes reconstructions truncated below $\sim 90\%$ determinant coverage.

Read alongside the independence figure, whose idealized full-state curve first fires near $\sim 7\%$ retained determinants and whose device-realizable shared-state form fires near $\sim 18\%$ on the same $L = 12$ sector ($d \approx 1.6 - 4.4 \times 10^4$ of 2.4×10^5 ; Figs. 7, 8), this $\sim 90\%$ figure reflects not a sharper test but a different one: those curves are the large $L = 12$ sector scored by a *relative* discrepancy $|\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ against a fixed fractional line, whereas the present forecast is the small $L = 6$ sector scored by an *absolute* shot-budget interval $z\delta_1$ tied to the `ibm_fez` budget—so these coverage percentages are set against incommensurate rulers on different-sized sectors (and, at $L = 12$, against two reference states), not one test yielding several answers. The direction of the residual uncertainty must be read correctly for a refutation-only instrument. Underestimating δ_k shrinks the threshold $z\delta_k$, which is *safe against missed errors* (false negatives—the acceptable direction for a necessary condition) but *anti-conservative against false refutations* (false positives, the cardinal error for a necessary-condition test). The forecast’s false-positive safety therefore rests on modeling the reference as exact-by-coverage ($|\hat{m}_1 - \bar{m}_1| \sim 10^{-15}$), so that no true claim sits near the threshold; away from full coverage the threshold should instead be set on $\text{Var}(\hat{m}_k - \bar{m}_k)$, for which $\text{Var}(O_k)$ is a proxy valid only when the reference is trusted. We state the sensitivity plainly, since at twice the shot floor or a 4% bias the first moment alone passes some still-truncated reconstructions.

That fragility is a property of a single moment. It has an accidental blind spot—one intermediate truncation whose misplaced weight leaves m_1 coincidentally intact within its interval—and the joint battery (m_0, m_1, m_2) with the Hankel–Stieltjes feasibility test closes *that* blind spot: the second moment, weighing the spectral tail the first averages over, refutes the same reconstruction well beyond its interval ($|\hat{m}_2 - \bar{m}_2| \approx 3.1$ against a threshold ≈ 1.8) and keeps refuting at the doubled bias budget where m_1 alone goes blind. We report this as closing the blind spot *exhibited here*, not as a proof that the joint battery is free of blind spots in general—no finite battery can be, since necessary conditions are not sufficient. Two boundaries frame this demonstration. The discrepancy is measured against the exact oracle $\hat{m}_k$, so it *calibrates* the interval where classical ground truth exists; the fully device-side version, propagating measured rather than assumed bias, is the deployment step. And where the finite-shot moment-constraint machinery of Refs. [41, 42] uses feasibility to *certify* that a static property holds, we use *infeasibility* to *refute* a reconstruction against constraints it was never fit to—the screen falsifies and never certifies—applied to a frequency-resolved spectral (Lehmann) measure rather than an expectation value or steady-state property. The classical spectral-moment sum rules themselves are textbook [2, 75]; what is new is only their use as reconstruction-independent, interval-valued refutation constraints on a hardware-computed spectral function.

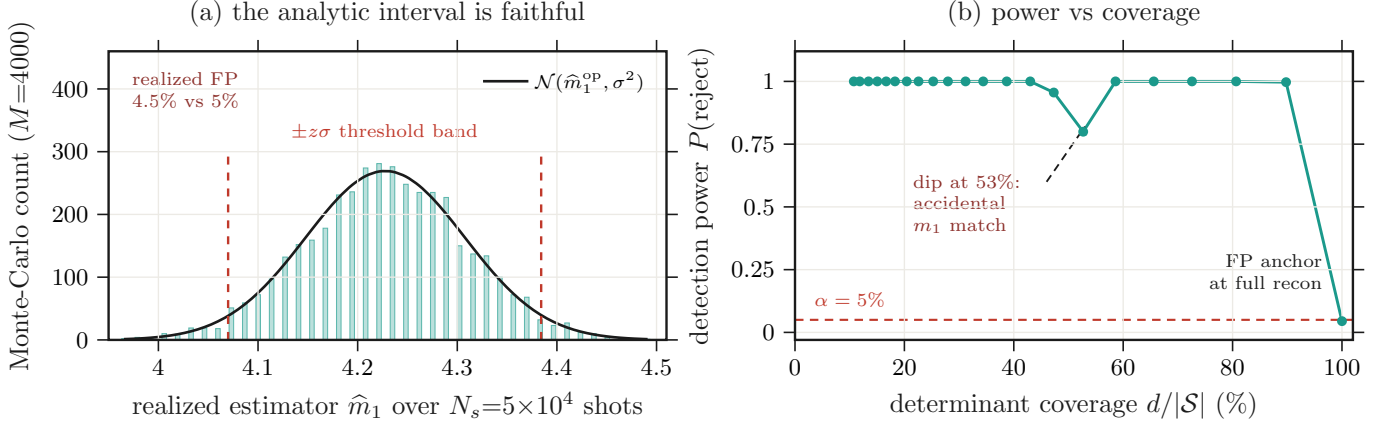


FIG. 14. **Direct Monte-Carlo demonstration of the interval-moment test at the `ibm_fez` budget** ($N_s = 5 \times 10^4$, $M = 4 \times 10^3$ replicas, current probe, $L = 6$, $U/t = 4$), computed sim-only by drawing shots from the exact-state distribution $|\langle x|0\rangle|^2$ —device bitstring counts were not retained, so this calibrates the interval at a classically reproducible scale and does not certify a device line shape. **(a)** The realized independent estimator $\hat{m}_1$ is Gaussian (skewness -0.04 , excess kurtosis -0.05), its spread matches the analytic shot rate to 1%, and its realized false-positive rate is 4.5% (Wilson-95% [3.9, 5.2]%) against the nominal 5%—validating the analytic interval rather than replacing it (the band shown is the shot-only interval $\pm z\sigma_{\text{shot}}$ under validation; the bias-inclusive deployment threshold $z\delta_1 \approx 0.23$ of the main text is wider). **(b)** Against synthetic truncated reconstructions the test reaches full detection power for coverage $\lesssim 90\%$, with a single honest exception near 53% coverage where an accidental first-moment match suppresses power; the (m_0, m_1, m_2) +Hankel–Stieltjes battery closes that blind spot exhibited here. The estimator variance is the exact-state local-estimator variance, a lower bound on the true per-shot uncertainty; a wider real interval only makes this refutation-only screen more conservative.

Appendix C: The Christoffel miss-distance: proof and evaluation

We prove the equality of the extremal cumulative-weight spread with the Christoffel function through three classical lemmas [67, 68, 70]. Write $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ for the Christoffel–Darboux kernel and $\lambda_n(t) = 1/K_n(t, t)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$ (for a finite N -atom Lehmann spectrum this holds for $n \leq N-1$).

Lemma 1 (variational identity). $\lambda_n(t) = \min\{\int |P|^2 d\mu : P \in \mathcal{P}_n, P(t) = 1\}$, with minimizer $P^* = K_n(\cdot, t)/K_n(t, t)$. Expanding $P = \sum_{j=0}^n c_j p_j$ gives $\int |P|^2 d\mu = \sum_j c_j^2$ while $1 = P(t)^2 = (\sum_j c_j p_j(t))^2 \leq (\sum_j c_j^2) K_n(t, t)$ by Cauchy–Schwarz, with equality at $c_j = p_j(t)/K_n(t, t)$; the minimum is $1/K_n(t, t) = \lambda_n(t)$.

Lemma 2 (maximal atom). Among positive measures ν with $\int \omega^k d\nu = m_k$ for $k \leq 2n$, the largest mass at a point t is $\max_\nu \nu(\{t\}) = \lambda_n(t)$. For the extremal P^* of Lemma 1, $\nu(\{t\}) = P^*(t)^2 \nu(\{t\}) \leq \int (P^*)^2 d\nu = \int (P^*)^2 d\mu = \lambda_n(t)$, the middle equality using $\deg(P^*)^2 = 2n$ and that ν shares those moments; the bound is attained by the $(n+1)$ -point Gauss–Radau rule with a node prescribed at t , a positive measure matching $m_0, \dots, m_{2n}$ whose weight at t is $\lambda_n(t)$.

Lemma 3 (ordinate gap). For t in the open interval $(\omega_{\min}, \omega_{\max})$ (the interior of the convex hull of $\text{supp } \mu$,

interior gap points included), one-sided polynomial majorant/minorant $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \leq \mathbf{1}_{(-\infty, t]} \leq Q_+$ and $\int (Q_+ - Q_-) d\mu = \lambda_n(t)$ bracket $F_\nu(t) = \int \mathbf{1}_{(-\infty, t]} d\nu$ between $\int Q_- d\mu$ and $\int Q_+ d\mu$ for every moment-matching ν ; the bracket width is $\lambda_n(t)$, attained by the upper and lower principal (Gauss–Radau) representations with a node fixed at t , read right- and left-continuously: these two moment-matching measures differ at t by exactly the atom $\lambda_n(t)$, and their existence is the classical principal-representation theorem [67, 68, 97, 98]. Hence $\sup_{\nu_1, \nu_2} |F_{\nu_1}(t) - F_{\nu_2}(t)| = \lambda_n(t)$.

Combining the three, the extremal spread equals $\lambda_n(t) = W_n(t)$, and for any single reconstruction consistent with the moments $|F_{\text{rec}}(t) - F_{\text{true}}(t)| \leq W_n(t)$ everywhere; only at or beyond the spectral edges $\omega_{\min}, \omega_{\max}$, where a single principal representation survives, does the equality weaken to this upper bound, whereas at interior gap points—where both principal representations exist—the equality persists. Monotonicity $W_{n+1} \leq W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \geq K_n$.

Evaluation of the certificate from the moments. The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short, numerically stable pipeline [70, 72]. With the inner product $\langle f, g \rangle_\mu = \int f(\omega)g(\omega) d\mu(\omega)$, so that $\langle \omega^a, \omega^b \rangle_\mu = m_{a+b}$, the orthonormal polynomials follow from Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ with $p_0 = 1/\sqrt{m_0}$ and obey the three-term recurrence

$$\beta_{j+1} p_{j+1}(\omega) = (\omega - \alpha_j) p_j(\omega) - \beta_j p_{j-1}(\omega), \quad (\text{C1})$$

with $\alpha_j = \langle \omega p_j, p_j \rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1} \rangle_\mu$, all fixed by $\{m_k\}$; this Stieltjes recurrence is numerically preferable to inverting the Hankel matrix directly. The nodes and weights of the $(n+1)$ -point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \text{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–Welsch), and the kernel on the diagonal, $K_n(t, t) = \sum_{j=0}^n p_j(t)^2$ evaluated by the same recurrence, yields the certificate $W_n(t) = 1/K_n(t, t)$. Every quantity in the guarantee is thus an explicit, auditable function of the measured moments. Because $K_n(t, t)$ is evaluated pointwise by the same recurrence, the certificate can be read at any frequency, including one in a weight-free region: on the $L = 6$, $U/t = 8$ current measure a 60-digit evaluation at a point in the empty high-frequency region gives $W_1(t^*) \approx 2.4 \times 10^{-3}$, $W_2(t^*) \approx 2 \times 10^{-5}$, and $W_n(t^*) < 10^{-5}$ for $n \geq 3$, against a global maximum $\max_t W_4 \approx 0.875$ —the bound is tight in the weight-free region where the global bound is loose.

Appendix D: Complementary state-level falsifier: the Gauss law

The Gauss-law check of Results I is complementary to the moment screen rather than a second instance of it, and we record its scope and construction here. In the $U(1)$ lattice Schwinger model the generator of gauge transformations G_x commutes with the Hamiltonian, $[H, G_x] = 0$ at every site x , and every physical state lies in the $G_x = 0$ sector, so $\langle \sum_x G_x^2 \rangle = 0$ exactly. Because $\sum_x G_x^2$ is a sum of local operators diagonal in the computational basis, it is estimated from the same samples at marginal cost and needs neither a reconstruction nor a moment. It is therefore a *state-level* check: it fires on gauge-symmetry breaking—leakage out of the physical sector—an error class the reconstruction moment screen does not target. Figure 15 shows the diagnostic on a small instance ($N = 4$ staggered sites, 3 spin- $\frac{1}{2}$ links, OBC, $w = 1$, $m = 0.6$): as the prepared state is contaminated by a fraction ϵ of unphysical weight, $\langle \sum_x G_x^2 \rangle$ rises monotonically from zero. Gauss-law violation as a measurable diagnostic was established experimentally in an analog cold-atom simulator [34, 118]; the digital lattice-Schwinger model is the setting of self-verifying variational simulation [37] and early digital gauge-theory dynamics [119, 120].

Appendix E: Within-sector redistributions: injectivity and the reach of the deployed battery

This appendix proves Theorem 2 and instantiates it numerically. The one boundary stated in Results II—that the independent estimator detects weight leaking *outside* the retained determinant support but not a redistribution strictly *within* it that preserves the moments used (Sec. IV)—is a genuine residual, characterized and bounded rather than fundamental. Its three ingredi-

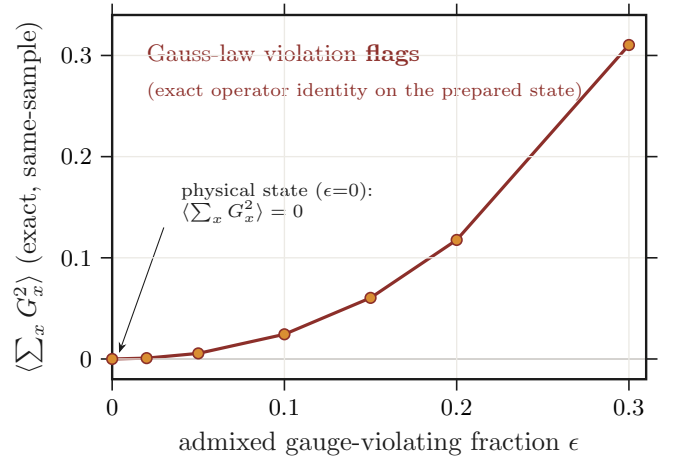


FIG. 15. **A complementary state-level falsifier: the exact Gauss-law constraint.** In the $U(1)$ lattice Schwinger model every physical state satisfies $\langle \sum_x G_x^2 \rangle = 0$ exactly, because $[H, G_x] = 0$. As the prepared state is contaminated by a fraction ϵ of unphysical (gauge-violating) weight, $\langle \sum_x G_x^2 \rangle$ rises monotonically from zero and the constraint is violated. This is a native, exact, same-sample operator identity evaluated directly on the state—a falsifier for gauge-symmetry breaking, a different error class from the weight/subspace errors the reconstruction moment screen targets, and by a different mechanism (a state-level check, not a moment screen on a reconstructed spectrum).

ents are established in turn: Vandermonde injectivity and the completeness bound Eq. (2) (*Injectivity*, proving part (a)); the super-resolution collapse of $\sigma_{\min}$ that converts exact injectivity into a shot-limited detectability threshold (*Conditioning*, proving part (b) and the Corollary); and the worst-case linear program against the deployed thresholds that instantiates both on the committed $L = 6$, $U/t = 4$ spectrum (*Reach*, Table III). We characterize the residual and quantify how far the deployed battery reaches into it; we do not claim to eliminate it.

Injectivity (proof of Theorem 2(a)). A $T = 0$ Lehmann current spectrum on a finite sector is a discrete measure $A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n)$ on N distinct poles $\{\omega_n\}$, with $w_n = |\langle n | J | 0 \rangle|^2 \geq 0$. Its moments are $m_k = \sum_n w_n \omega_n^k$, i.e. $\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ a Vandermonde matrix. Because the poles are distinct, V restricted to rows $0, \dots, N-1$ is invertible, so the weight-to-moment map is injective through order $N-1$: a nonzero within-sector redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \neq 0$ cannot preserve all of $m_0, \dots, m_{N-1}$. Hence any redistribution preserving $m_0, \dots, m_K$ ($K < N-1$) is refuted by *some* higher moment m_j with $K < j \leq N-1$ (and $j \leq 2N-1$ if the retained poles are also allowed to move, by Prony counting). The blind spot is thus not the failure of the construction but a statement about which moment order resolves it. For the doped Hubbard current spectrum of Results II ($L = 6$, $U/t = 4$) the current-active eigenstates fall into $N = 5$ distinct excitation

TABLE III. Within-sector redistributions of the $L = 6$, $U/t = 4$ current spectrum ($N = 5$ distinct poles), constructed in the null space of $[\omega^0, \dots, \omega^K]$ to preserve $m_0, \dots, m_K$ exactly. “Worst-case Δ_{K+1} ” is the feasible range from the linear program Eq. (E1); “deployed threshold τ_{K+1} ” is the 95% two-sided falsification threshold $\tau_k = z_{0.975} \delta_k$ at the `ibm_fez` budget (App. B, $z_{0.975} = 1.96$; note τ already includes the factor z , distinguishing it from the 1σ half-width δ of App. B), with “–” where that moment order exceeds the deployed battery. A redistribution is refuted only when its induced shift on some deployed moment exceeds the corresponding τ .

preserved through	first non-zero moment	worst-case Δ_{K+1} (LP, Eq. (E1))	deployed threshold τ_{K+1}	in deployed battery?
m_0, m_1	m_2	$[-0.92, 7.09]$	1.848	yes: refuted if large
m_0, m_1, m_2	m_3	$[-0.14, 5.26]$	–	no: exact-only tooth
m_0, m_1, m_2, m_3	m_4	$[-0.30, 0.39]$	–	no: exact-only tooth

frequencies (sixteen nonzero-weight eigenstates grouped into five degenerate frequencies), so the Lehmann measure has five atoms and the map is injective through m_4 : any within-sector redistribution among the five distinct poles preserving $m_0, \dots, m_K$ is refuted by some m_j with $K < j \leq 4$ (at most m_9 if the poles also move). Weight traded among the degenerate eigenstates at a single frequency changes no moment and leaves $A(\omega)$ unchanged—it is not an error but the exactly-degenerate trading noted below. The practical consequence is favorable: the residual is resolved already at m_4 , within the low-order reach of the screen rather than at some high inaccessible order.

Conditioning: injectivity is not resolvability (proof of Theorem 2(b)). The invertibility of V on rows $0, \dots, N-1$ is a statement in exact arithmetic: it guarantees a unique preimage $\mathbf{w} = V_N^{-1} \mathbf{m}$ on the $N \times N$ node-Vandermonde V_N , but it does not control how a finite-shot moment estimate resolves that preimage. The amplification of moment error into weight error is set by the condition number $\kappa(V_N) = \sigma_{\max}/\sigma_{\min}$, and the operative obstruction is that $\sigma_{\min}(V_N)$ collapses when poles cluster. In the sub-Rayleigh (super-resolution) regime, where s poles lie within a separation η small against the spectral scale, the smallest singular value decays like a power of the separation, $\sigma_{\min}(V_N) \sim \eta^{s-1}$ —the sharp Vandermonde-conditioning phase transition of Moitra and the clustered-node bounds of Batenkov *et al.*, the moment-domain face of the classical super-resolution limit [100–103]. Consequently “injective through m_{N-1} ” overstates what is operationally resolvable: a within-sector redistribution supported on clustered poles perturbs the exact moments by an amount that can fall below the shot-noise floor δ_k , so it is formally distinguished by m_{N-1} yet unresolvable at any shot-affordable order. The honest criterion is therefore not the exact-arithmetic order $N-1$ but the smallest j at which the induced moment shift clears the threshold: a redistribution is resolved at m_j *provided* the pole-cluster conditioning—equivalently $\sigma_{\min}$ of the relevant Vandermonde block, or the minimal pole separation relative to the super-resolution limit fixed by the shot budget—keeps $\Delta_j^{\max}$ of Eq. (E1) above τ_j . This is exactly what the linear program Eq. (E1) and Table III measure; the conditioning framing supplies the reason a formally exposable shift can be numerically negligible—clustered poles—

and identifies the residual as a super-resolution obstruction [100, 102], not a defect of the injectivity statement. For the $L = 6$, $U/t = 4$ current spectrum treated here the $N = 5$ poles are well separated (the induced shifts of Table III are $O(1)$, far above the thresholds τ_k), so conditioning is benign in this instance; the caveat bites only when the sampled sector carries near-degenerate excitation frequencies, where no shot-affordable moment order resolves a redistribution among the cluster.

Reach of the deployed battery (instantiation of Theorem 2(b) and Corollary). Injectivity in exact arithmetic is not the operational question; the deployed screen uses only the shot-affordable moments $\{m_0, m_1, m_2\}$ with the 95% falsification thresholds $\tau_k = z_{0.975} \delta_k$ of App. B ($\tau_0=0.029$, $\tau_1=0.229$, $\tau_2=1.848$ at the `ibm_fez` budget and 2% readout bias; τ includes the factor $z_{0.975} = 1.96$, unlike the 1σ half-width δ). For a redistribution that preserves $m_0, \dots, m_K$, the largest shift it can impose on the next moment is the linear program

$$\Delta_{K+1}^{\max} = \max_{\delta} \left| \sum_n \delta_n \omega_n^{K+1} \right| \quad \text{s.t.} \quad \sum_n \delta_n \omega_n^j = 0 \quad (j \leq K), \quad \delta_n \geq -w_n, \quad (\text{E1})$$

the constraint $\delta_n \geq -w_n$ keeping the corrupted weights physical. Table III reports Eq. (E1) against the deployed thresholds. A redistribution preserving only m_0, m_1 can move m_2 by as much as $\Delta_2^{\max} \approx 7.1$, far above $\tau_2 = 1.848$: the deployed (m_0, m_1, m_2) battery therefore refutes every redistribution whose induced shift on a deployed moment exceeds its threshold τ_k , and the showcase near-instance on the same $L = 6$, $U/t = 4$ interval-battery system (retained subspace dimension $d=98$, where $|g_1| = 0.225 < \tau_1$ but $|g_2| = 3.14 > \tau_2$) is exactly a within-band case the joint battery still rejects. The complementary caveat must be stated plainly: because the criterion is $|\Delta_m| > \tau_k$, a redistribution that keeps every deployed moment within its threshold survives regardless of how much of the line shape it alters—a fine-tuned within-band corruption can move a large fraction of the weight while shifting m_0, m_1, m_2 only slightly. The irreducible residual is exactly this set: a redistribution engineered to keep m_0, m_1, m_2 within tolerance, whose first exposable signature is at m_3 (in exact arithmetic $\Delta_3^{\max} \approx 5.3$, but m_3 lies beyond the reliably estimable order at this budget), together with weight traded among poles unresolved

within the sampling interval and exactly-degenerate trading (which leaves $A(\omega)$ unchanged and is not an error). We disclose this fine-tuned residual rather than claim to close it; positivity (Hankel/Stieltjes) is blind to it by con-

struction, since a within-sector redistribution is itself a valid positive measure, so only an independent higher moment—not the geometry—resolves it.

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